

Review

Grasping emergency dynamics: A review of group evacuation techniques and strategies in major emergencies

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ABSTRACT

Major sudden disasters, such as floods, earthquakes, and fires, often cause significant casualties. Emergency evacuation is crucial in mitigating these impacts. Different types of disaster incidents vary significantly in terms of impact scope, suddenness, and urgency. Each type of disaster possesses distinct characteristics, necessitating varying requirements for emergency evacuation. Consequently, we conducted a bibliometric analysis and visual mapping of evacuation processes in major natural disasters from 2004–2023, analyzing 7213 publications from the Web of Science database via VOSviewer and ArcGIS. Our study identified three developmental phases: an initial phase (pre-2011) with 1169 publications, a growth phase (2012–2018) with 2772 publications, and an expansion phase (post-2019) with 3335 publications. This study provides a comprehensive review and classification of emergency evacuation theories and methods in major disaster scenarios. It emphasizes the necessity of assessing the scope and intensity of different types of major emergent disasters, defining and simulating the affected behaviors of the influenced populations, and formulating differentiated emergency evacuation strategies accordingly. Keyword analysis reveals two main trends supporting these findings: an increasing focus on complex evacuation modeling and simulation techniques, manifested in the application of various simulation-optimized microscopic and macroscopic models such as cellular automata, social force models, agent-based models, pedestrian flow, and network flow models, enhancing disaster understanding and prediction capabilities; and the strategic development of tailored evacuation strategies for specific disaster contexts, thereby improving disaster response efficiency. Three key future pathways for safety evacuation research are outlined: refining evacuation behavior models for greater accuracy, improving the coordination of complex, multi-level evacuation procedures, and integrating indoor and outdoor evacuation strategies more seamlessly. It establishes a forward-looking framework for advancing safety evacuation studies in major emergencies.

1. Introduction

Over the past two decades, global emergencies, including floods, fires, and earthquakes, have posed significant challenges to mass evacuation and crisis management. According to EM-DAT, 12,244 such disasters were documented worldwide from 2004 to 2023 [1]. Concurrent socio-economic growth has resulted in an increase in densely populated, complex structures such as high-rise buildings and sports arenas, which have complicated evacuation efforts during major incidents such as stampedes and natural disasters, frequently leading to high casualties and property damage. Consequently, efficient emergency evacuation has become a critical focus. This involves **minimizing casualties,**

preventing secondary incidents, reducing overcrowding, providing prompt medical aid, and safeguarding infrastructure. In 2022 alone, seven significant earthquakes resulted in over 10 fatalities each, with three exceeding 100 deaths [1]. In 2022, Seoul Itaewon stampede resulted in over 300 casualties [2]. Therefore, **the development of effective group evacuation models, the understanding of crowd behaviors, and the establishment of preventive strategies are increasingly vital in emergency response planning.**

Evacuation from large-scale natural meteorological disasters such as typhoons and floods has benefited from advanced satellite weather forecasting, which provides people with more time to prepare and evacuate from low-lying areas to high ground. Earthquakes necessitate

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an immediate and large-scale response due to their abrupt onset and extensive impacts, which places significant demands on the speed and efficiency of emergency response. Building fires are usually limited to individual structures; however, due to the rapid spread of fire, quick evacuation is necessary to avoid entrapment or exposure to smoke and toxic gases. Globally, the research approaches vary: Liu et al. [3] proposed an emergency evacuation method for high-density communities that considers the impact of building debris, whereas Li et al. [4] proposed a disaster assessment method that can identify floods and enable people to evacuate along the shortest path possible. In urban scenarios with unforeseen accidents, Ahn and Choi [5] provided the safest evacuation route considering fire spread, and Deng et al. [6] considered different ranges of CO concentration, temperature, and smoke visibility to achieve the fastest evacuation on a cruise ship. For other disasters, Khan et al. [7] integrated machine learning tools and GIS to quantify the risk in a region. A significant gap in the existing literature is the lack of comprehensive studies that specifically address the evolving characteristics of each type of disaster. Targeted simulation and evacuation strategies tailored to the unique nature of each disaster, such as dynamic fluctuations, extensive reach, and varying intensity levels, are needed.

In this study, floods, earthquakes, and building fires are examined as distinct emergency examples to enhance preparedness and manage future crises effectively. Through the analysis of various disaster evacuation procedures, two main objectives are addressed: improving evacuation models based on the impacts of different disasters' scopes, intensities, and dynamic characteristics on pedestrian behaviors, and formulating strategies to manage different disaster scenarios. This approach aims to enhance evacuation effectiveness through a comprehensive review and analysis of current methodologies, including evacuation simulation, model refinement, and strategy optimization. The consolidated analysis aims to provide a clear and concise understanding of emergency evacuation practices and their efficacy in managing large-scale emergencies.

2. Methodology and background in the literature review

2.1. Literature collection

This study conducts a systematic literature review spanning from 2004 to 2023, focusing on evacuation processes during disasters. The Web of Science Core Collection was utilized to employ keywords such as (evacuation or evacuate or evacuees) AND (disaster or catastrophe or hazard or calamity or earthquake or flood or fire) AND (modeling or

simulation or discrete model or continuous model or microscopic model or macroscopic model or optimization) AND (strategy or approach or method or procedure or technique or plan or tactic or path planning) to filter pertinent publications from the last two decades. The selection criteria for document types were "article" and "review article", and the time span was set to 01/01/2004-31/12/2023. All of these enabled the analysis of literature specifically relevant to disaster evacuation strategies. VOSviewer software was employed to visually map and preliminarily assess the field's research contributions, aiming to identify trends, hotspots, and interconnections among various topics in the context of disaster-induced evacuations. The methodological approach not only helps in revealing potential future research directions but also in understanding the nuances of evacuation processes in disaster scenarios. Additionally, the study includes a detailed flowchart, shown in Fig. 1, illustrating key steps in the disaster evacuation process and providing a structured framework for the ensuing discussion.

2.2. Literature background and classification

2.2.1. Temporal distribution of research activities

In the realm of scientific literature, publications during the initial phase (Pre-2011) were relatively rare, with 1169 publications, reflecting the early stages of development in this field. However, the study of group evacuation has witnessed a notable escalation, marked by a 15% annual growth between 2008 and 2010, as a societal response to events like the Wenchuan earthquake [8]. Advancements in numerical simulations have further fueled research intensity and innovation. A surge of 2772 publications during the identified growth phase (2012-2018) marked a significant increase in research efforts. As illustrated in Fig. 2, evacuation research has evolved through a trajectory of consistent exploration and growth in recent years. This evolution has been significantly driven by technological advancements, including the use of big data and the Internet of Things (IoT) [9-12], leading to what has been identified as the expansion phase (Post-2019). The most recent phase, with 3355 publications, highlights the field's burgeoning prominence, underscored by its critical role in managing disasters. The number of publications at this stage continued to increase and tended to stabilize, demonstrating the in-depth application and innovation in disaster evacuation. Based on the first two stages, researchers further explored and applied the theory and practice of disaster evacuation, proposing more solutions and innovative methods, and important research achievements and practical cases emerged. Overall, this analysis delineates the evolving landscape of evacuation research, mapping its

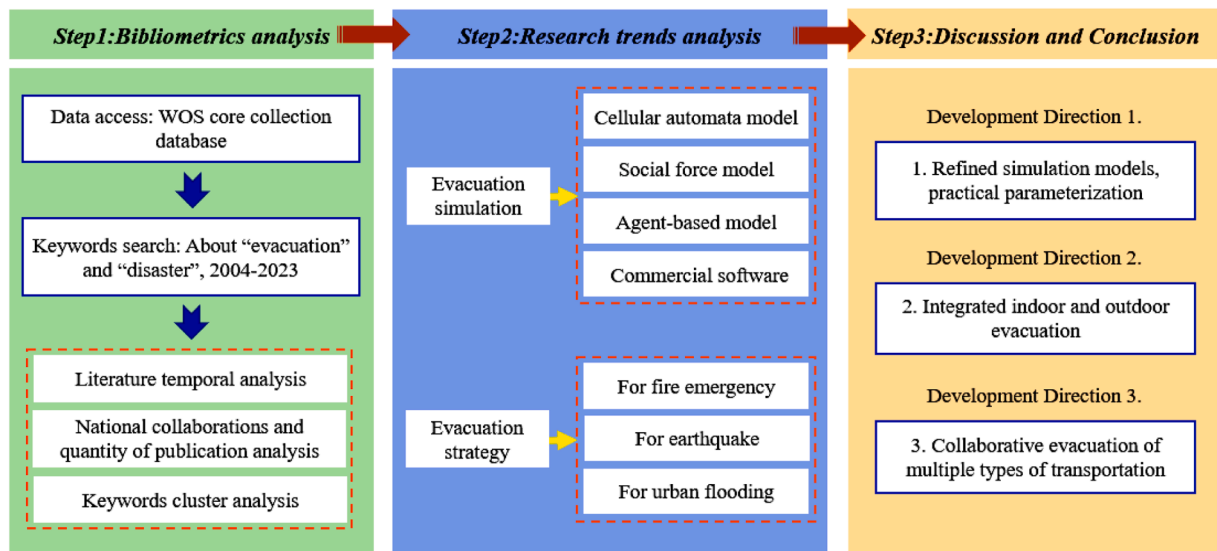


Fig. 1. General structural outline of the paper.

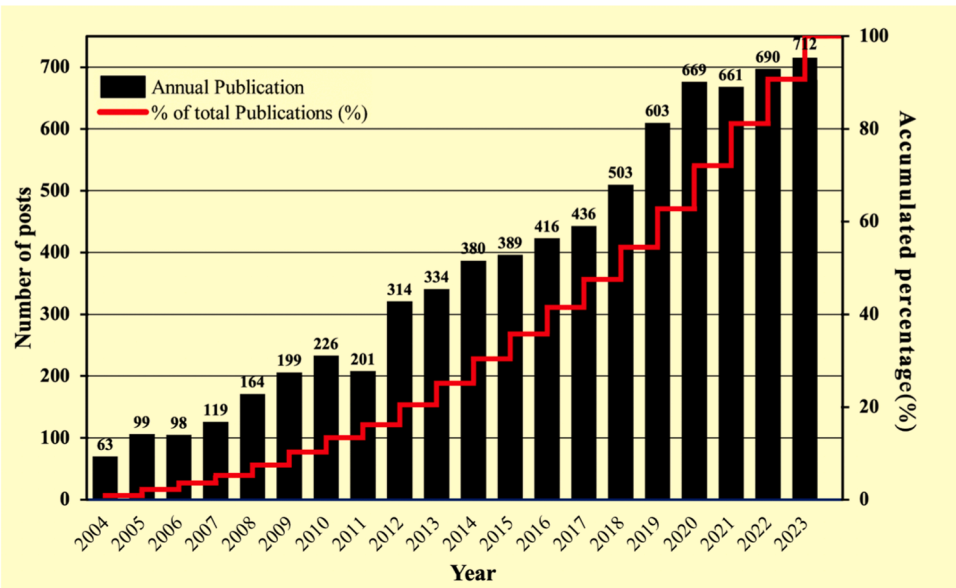


Fig. 2. Temporal distribution of literature as of December 31, 2023.

progression from foundational studies to its current state, where advanced technologies are integrated to address complex real-world challenges.

2.2.2. Patterns and implications of state-to-state collaborations

A collaborative map among the top 35 countries in evacuation research is presented, as depicted in Fig. 3. The countries are represented by circles, sized by publication count, with connecting lines indicating collaborative intensity based on joint publications. China (1958 publications) and the United States (1937 publications) lead in volume, reflecting their diverse geographies and frequent natural disasters which demand extensive research and collaboration. Other notable contributors include Japan, the United Kingdom, Australia, Italy, Canada, and Germany, highlighting the global nature of this research area. This map underscores the importance of international collaboration in addressing disaster and evacuation challenges, a trend expected to grow in significance.

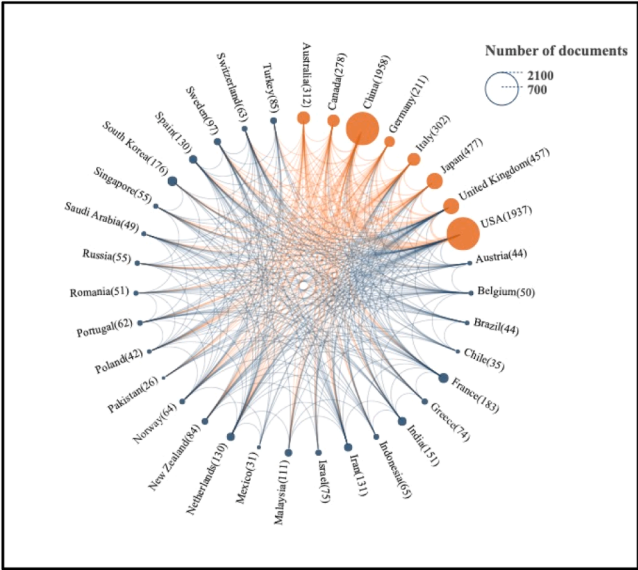


Fig. 3. Collaborative map of the top 35 countries with the highest number of publications on evacuation research.

2.2.3. Keywords and research trends analysis

In this study, the selected literature from the Web of Science Core Collection was exported in plain text format and imported into VOSviewer bibliometric software. The "Minimum number of occurrences of a keyword" was set to 25, and keywords with repetitive meanings were merged, such as "evacuation" and "evacuate", "disaster intensity" and "disaster duration", "crowd" and "evacuee" and "pedestrian", "strategies" and "strategy", and "dynamics" and "dynamic". Table 1 shows the ten most frequently used keywords. These keywords are all derived from the literature related to disaster evacuation; therefore, "evacuation" and "emergency" have received high reviews. The increasing frequency of natural and artificial disasters has highlighted the importance of disaster evacuation, leading to a greater focus on public safety and evacuation protocols. Keywords such as "simulation" and "model", as important methods and tools of evacuation, are featured prominently, indicating the use of computer simulation to characterize the movement patterns of evacuated pedestrians. "Disaster characteristics" and "dynamic" indicate attention to disaster characteristics and the dynamic evolution of disasters in terms of scale, intensity, and other factors. The high-frequency terms "behavior" and "crowd" reflect the increasing focus on pedestrian evacuation behaviors and the development of more refined crowd behavior models. "Strategy" and "optimization" demonstrate the emphasis on optimizing evacuation strategies. Overall, the table reflects the focus and research on disaster characteristics, crowd evacuation models in emergencies, and optimized evacuation strategies, which are crucial aspects of disaster evacuation research.

In the domain of evacuation research, a keyword co-occurrence network map, as depicted in Fig. 4, identifies four key clusters: 1) The

Table 1
Quantitative analysis of the most prevalent 10 keywords

Rank	Frequency	Keyword
1	1391	evacuation
2	887	model
3	883	emergency
4	819	disaster characteristics
5	763	simulation
6	655	behavior
7	522	crowd
8	471	strategy
9	368	dynamic
10	359	optimization

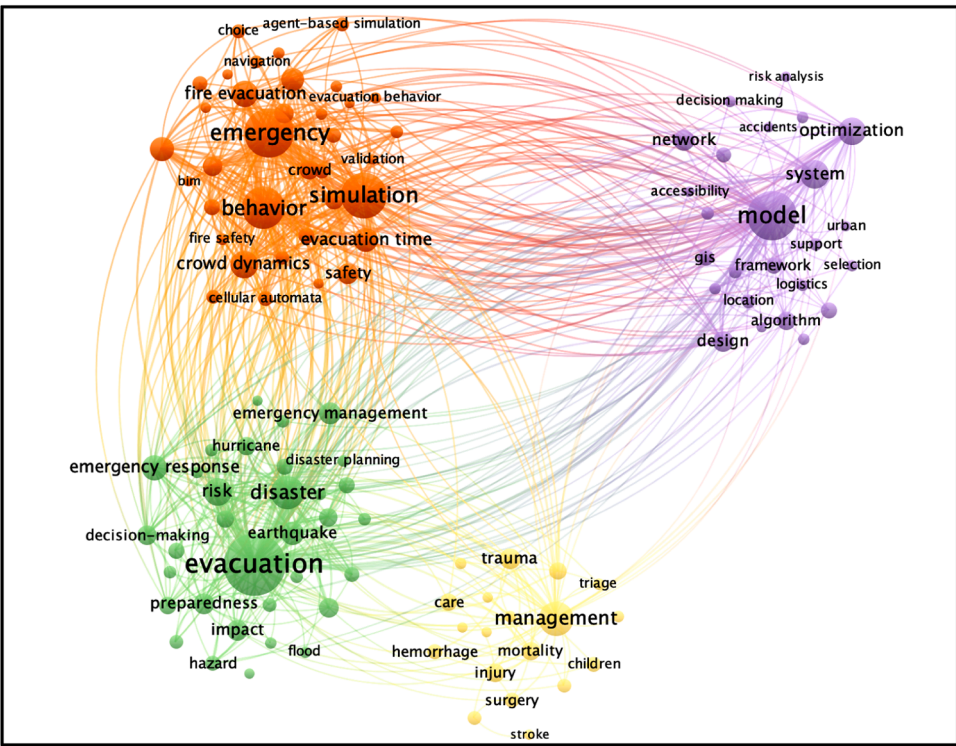


Fig. 4. Mapping of the co-occurrence of keywords in disaster evacuation research.

Simulation Cluster centers on "simulation", "emergency", and "behavior", delving into evacuation technologies and human conduct in emergencies. 2) The Risk Assessment Cluster, characterized by "evacuation", "disaster", and "risk", emphasizes risk evaluation, decision-making, and post-disaster recovery strategies. 3) The Optimization Cluster, featuring "model", "optimization", and "system", focuses on the development and enhancement of evacuation models and systems. 4) The Management and Care Cluster, revolving around "management" and "care", addresses

disaster preparedness and comprehensive care in evacuations. These clusters collectively underscore the research focus on simulation studies, behavioral analysis in emergencies, risk management, model optimization, and inclusive evacuation strategies. To elucidate emergency evacuation procedures, an analysis has been conducted on the most-cited papers in this domain, with key findings summarized in Table 2. These studies encompass a diverse range of evacuation contexts, including hurricanes [13,14], maritime incidents

Table 2
Top 10 most cited papers in the field

Cited number	Ref.	Title	Journal	Impact factor (2022)	Research issues
347	Rocklov et al. [19]	COVID-19 outbreak on the Diamond Princess cruise ship: estimating the epidemic potential and effectiveness of public health countermeasures	Journal of Travel Medicine	25.7	COVID-19; Scenarios of earlier versus later ship evacuation
304	Murray-Tuite and Wolshon [13]	Evacuation transportation modeling: An overview of research, development, and practice	Transportation Research Part C-Emerging Technologies	8.3	Evacuation planning; Hurricane; Evacuation travel demand; Infrastructure use; Management strategies
241	Huang et al. [14]	Who Leaves and Who Stays? A Review and Statistical Meta-Analysis of Hurricane Evacuation Studies	Environment and Behavior	5.7	Hurricane evacuation; Statistical meta-analysis; Actual evacuations; Hypothetical scenarios; Hazard warnings
226	Tomer et al. [20]	Socializing in emergencies-A review of the use of social media in emergency situations	International Journal of Information Management	21	Communication in emergency events; Social media tools
180	Liu et al. [18]	A path planning approach for crowd evacuation in buildings based on improved artificial bee colony algorithm	Applied Soft Computing	8.7	Path planning approach; Emergency evacuation simulation; Social force model; Artificial bee colony algorithm
167	Haghani and Sarvi [21]	Crowd behavior and motion: Empirical methods	Transportation Research Part B-Methodological	6.8	Crowd disasters; Emergency scenarios; Evacuation drills
138	Thompson et al. [22]	Evacuation from Natural Disasters: A Systematic Review of the Literature	Risk Analysis	3.8	Evacuation Behavior; Natural disaster; Risk Perception
107	Chen et al. [16]	Application of the best evacuation model of deep learning in the design of public structures	Image and Vision Computing	4.7	Evacuation design simulation for subway station; Deep learning; VR video tracking method; Evacuation behavior
70	Kang et al. [15]	An improved social force model for pedestrian dynamics in shipwrecks	Applied Mathematics and Computation	4	Evacuation in Shipwrecks; Social force model; Evacuation efficiency; Wicked pedestrians; Evacuation dynamics
45	Yang et al. [17]	Guide optimization in pedestrian emergency evacuation	Applied Mathematics and Computation	4	Social force model; Pedestrian dynamics; Guide assignment; Visual field

[15], and subway evacuations during natural disasters [16]. Methodologically, these papers employ approaches ranging from statistical meta-analysis [14], which aggregates real evacuation data, to the development of simulation models [17] and the application of social force models for detailed crowd dynamics [18]. The research scope of these papers is broad, addressing aspects such as evacuation demand assessment, infrastructure capacity, the influence of hazard warnings on evacuation efficacy, the role of social media in emergency communication, path planning, crowd simulation in evacuation scenarios, and factors influencing evacuation efficiency. Notably, one study [19] introduces a novel perspective by examining evacuation in the context of the COVID-19 pandemic. Collectively, these papers provide comprehensive insights into the dynamics of evacuation in various emergencies, contributing both theoretically and practically to the field.

The study of disaster evacuation, still in an exploratory phase, has shown immense potential for advancement. The analysis of keyword frequency and centrality has identified two pivotal research trends shaping the field: evacuation modeling and simulation and disaster evacuation strategies. The first trend emphasizes the role of evacuation simulation technology as a vital tool for understanding and predicting evacuations in diverse disaster scenarios. It involves exploring evacuation models, evacuation tools, and simulations of human behaviors in the dynamic evolution of disasters with varying scales and intensities to understand evacuee crowd dynamics intricately and holistic evacuation procedures. The second trend focuses on formulating and optimizing evacuation strategies tailored to different types of disasters. Both trends have garnered significant attention and are expected to continue influencing the field.

3. Methods, techniques, tools and strategies

This section offers a focused analysis of current emergency evacuation theories and methods in disaster scenarios, emphasizing simulation methods for evacuations under disaster conditions and the optimization of evacuation strategies to provide comprehensive insights into this evolving field.

3.1. Evacuation models

Computer simulation has been pivotal in studying major emergency evacuations since the 1980s [23]. According to the modeling scale, evacuation models can be divided into macroscopic models and microscopic models. Importantly, both microscopic and macroscopic models have distinct strengths and limitations. The microscopic models focus more on modeling and simulating the behaviors of individual evacuees. The goal of macroscopic models is to optimize the problem by considering the global behavior and flow of evacuees to provide the best evacuation strategies and routes. In other words, the crucial difference between the two approaches is that macroscopic models focus on the overall crowd dynamics and flow characteristics, while microscopic models excel in capturing individual pedestrian behaviors and local interactions.

3.1.1. Microscopic models

When studying individual movement behaviors in major disaster situations, microscopic models are frequently employed. These simulations replicate individual movements within building structures to calculate evacuation times and visually display the evacuation process [24]. Recent advancements have integrated individual characteristics [25], interpersonal dynamics [26], and complex environmental factors [27]. Key microscopic models such as cellular automaton [28], social force [29], and multi-agent systems [30] are instrumental in this field. As they evolve, these models increasingly enhance emergency planning by offering detailed insights into crowd dynamics during evacuations. Compared to conventional evacuation models, models under disaster scenarios need to focus not only on evacuation within static building

scenes but also on considering dynamic changes and response measures in the surrounding environment [31]. These contribute to the increased complexity in the establishment of evacuation models. Moreover, evacuation models in major disaster scenarios also need to account for irrational behaviors affected by the disaster environment, such as panic [32], negative emotion [33], and congestion [34].

1) Cellular Automata

The cellular automata model, renowned for its efficiency and scalability [35], is extensively utilized in evacuation simulations. Originating from Von Neumann's concept [36], it operates on a grid where cells evolve according to specific rules and interactions with adjacent cells. This model relies on evolution rules, neighborhood structures, and cell states [37], with typical representations of cell neighbors depicted in two-dimensional spaces like Fig. 5.

Based on the coupling mechanism, the evacuation cellular automata model is categorized into a lattice gas model, floor field model, and other field-based models [28]. Muramatsu et al. [38] proposed a lattice gas model, known as a drift random walk model, where individuals are treated as particles within the cellular automata framework. Each individual is characterized by a preferred direction at each moment, and the movement of the individual is modeled via drift probabilities. Burstedde et al. [39] first proposed a floor field model based on a two-dimensional cellular automata framework. They simulated pedestrian flow, employed the concept of "field" to elucidate some typical phenomena, such as "queuing", and successfully reproduced the classic self-organizing phenomena. As shown in Fig. 5, the static floor field typically considers the building structure and layout, whereas the dynamic floor field takes into account dynamic changes in human behavior and environmental factors such as smoke.

In cellular automaton models, individuals primarily make rational choices for their next actions based on their perception of the surrounding environment [40]. However, basic cellular automaton models struggle to explicate complex crowd evacuation processes. To ensure simulation results more closely resemble real evacuations, scholars have continuously developed diverse cellular automaton models tailored to different application scenarios, as listed in Table 3.

Cellular automaton (CA) models, especially the floor field model, are widely used in evacuation simulation research because of their capacity to replicate individual decision-making influenced by environmental perception [56]. However, while these models excel in simulating human behaviors and evacuation processes [57], they are limited in their ability to capture the full spectrum of human complexity and nuanced interactions [58,59], as observed in experimental settings such as classrooms, restaurants and hospitals. Lattice gas models, though simpler and more applicable to geometrically straightforward scenarios [28], are inadequate in portraying complex human behaviors and evacuation strategies and are less scalable compared to floor field models. Despite their utility in certain contexts, the floor field model is distinguished by its more comprehensive approach. Improving the model and adjusting its parameters enables more accurate simulation of crowd behaviors in emergency situations and facilitates more targeted evacuation strategies. The continuous optimization and refinement of cellular automata models are essential for achieving more accurate and reliable simulations of major disaster evacuation.

Theoretical research and these optimized results provide a more precise and reliable tool for evaluating evacuation strategies, predicting personnel flow, and optimizing emergency planning in major disaster scenarios. Furthermore, combining these optimization results with disaster dynamics, range, and strength characteristics allows for a better response to major disasters. Hou et al. [60] used the floor field model to simulate the evacuation process of a large fire in Jilin, China and studied pedestrian exit selection behaviors. Considering pedestrians' subjective perceptions and collisions during earthquakes, Song et al. [61] simulated the evacuation process of an entire building. Sun et al. [62]

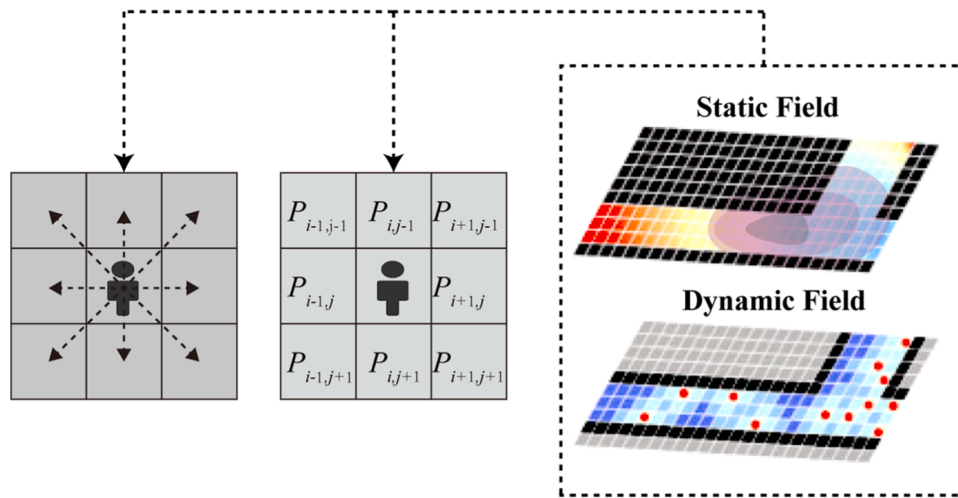


Fig. 5. Schematic diagram of floor field cellular automata.

quantitatively described panic behavior and familiarity with exits in earthquake emergencies and conducted large-scale community pedestrian evacuation simulations. Yi et al. [63] elaborated on the event floor field and rolling factors in the floor field model to describe the impact of stampede events, studied pedestrian evacuation dynamics during stampedes, and developed evacuation strategies for stampede events. Du et al. [64] used fuzzy logic theory to quantify pedestrians' evacuation intentions in extreme rainfall situations, adjusting their decisions in real-time based on the received information. When personnel had a high evacuation intention, they were more inclined to evacuate; when their willingness to evacuate was low, they tended to wait for rescue or interrupt the evacuation process. Song et al. [65] established a cellular automaton model for cabin evacuation, introducing disaster level to describe passenger anxiety. Cao et al. [66] combined cellular automaton models with fluid dynamics models to quantify the effects of water depth, flow velocity, and flow direction on pedestrian movement speed at subway stations.

The strengths of cellular automata include simplicity, computational efficiency, and scalability. The scalability allows cellular automaton models to adapt to various scenarios, hazards, and individual characteristics, rendering them widely applicable in major disaster evacuation situations. Cellular automata models in major disasters typically consider the psychological and behavioral characteristics of individuals during disasters, thereby realistically reflecting personnel evacuation behavior in disaster scenarios.

2) Social Force

Introduced by Helbing [67], the social force model employs continuous motion equations to precisely track the dynamic positions of pedestrians. It incorporates interaction forces among pedestrians, interactions between pedestrians and the environment, and exit-directed forces, as depicted in Fig. 6 [68]. This model effectively replicates evacuation phenomena such as "fast is slow" [68] and "self-organization" [69], making it an optimal tool for modeling pedestrian evacuation under pressure. The social force model, a micro-scale approach, has been widely adopted in recent evacuation simulations and is recognized for its precision in depicting realistic crowd behaviors.

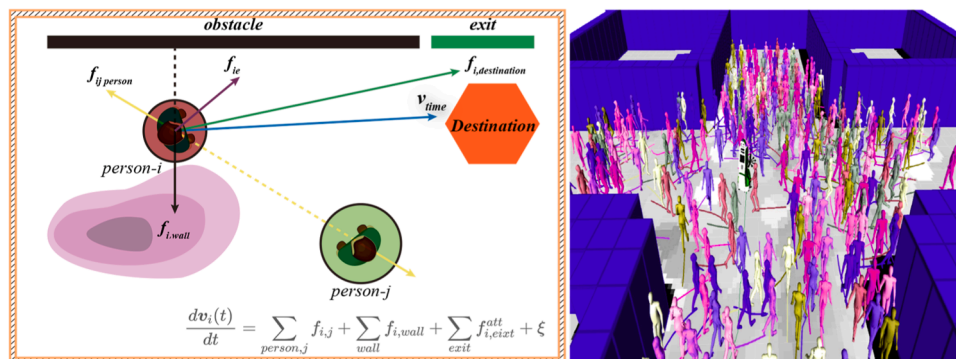
Advancements in evacuation research have necessitated modifications to the basic social force model to enhance its accuracy. Helbing et al. [69] examined pedestrian velocity changes as a reflection of emotional states, while Seyfried et al. [70] linked individual interaction range coefficients with desired interpersonal distances, resulting in velocity reduction in close proximities. Parisi et al. [71] introduced an automatic deceleration mechanism into the model, incorporating

clearance distance considerations for individual interaction ranges. Further refinements include the work of Frank and Dorso [72], who examined the effects of indoor structures on evacuation patterns, and Saboia and Goldenstein [73] who adapted the model with a moving grid for smoother trajectory management in congested areas. Gao et al. [74] integrated stress factors for more realistic disaster evacuation simulations, whereas Köster et al. [75] developed a simplified social force model to address oscillations and instabilities. Alia et al. [76] introduced a harmony search-based technique, utilizing global best particles from particle swarm algorithms to improve problem-solving efficiency and reduce computational complexity. These enhancements collectively advance the utility of social force model in simulating complex evacuation scenarios with increased accuracy and computational efficiency. As evacuation research diversifies and addresses its growing complexity, significant advancements have been made. Yang et al. [77] utilized ant colony algorithms in a pedestrian flow allocation model, thereby enhancing the social force model to predict individual behavior. Liu et al. [18] incorporated a visual impact factor, considering the influence of perspective on movement. Ibrahim et al. [78] used the social force model as the foundation for simulating basic pedestrian movement, incorporating member functions such as distance from the exit, familiarity and visibility of the exit, and crowd density around the exit into a fuzzy logic system to simulate evacuation under uncertain conditions in continuous space. Stuart et al. [79] examined interactions involving disabled individuals and integrated potential fields into the social force model to enable more nuanced pedestrian interaction simulations. Sticco et al. [80] aligned the outputs of the social force model with empirical pedestrian movement data by adjusting the friction coefficient. Han et al. [81] created a macro-micro optimization framework using genetic algorithms, focusing on factors such as exit width and evacuation density. Wei et al. [82] improved the pedestrian self-propulsion direction in the model and employed the particle pigeon swarm algorithm for multi-exit evacuation efficiency. Liu et al. [83] introduced ventilation force parameters, thereby enhancing the model's accuracy in depicting pedestrian behavior. Lin et al. [25] considered pedestrian characteristics and leadership roles, further refining the simulation of evacuation processes. Mao et al. [84] integrated the butterfly algorithm for emergency evacuation path planning, thereby improving resource utilization in multi-exit scenarios. García et al. [85] conducted a detailed parameter sensitivity analysis, providing an optimal parameter set for replicating both the macroscopic and microscopic aspects of pedestrian dynamics. The social force model has been modeled and validated by combining multiple optimization algorithms and empirical data to improve its predictive performance and practicality. These advancements collectively enhance the social force model's

Table 3

Key papers on cellular automata theory and applications.

No	Ref.	Model	Scenario	Innovations	Main influence Factors
1	Simulation of evacuation processes using a bionics-inspired cellular automaton model for pedestrian dynamics [41]	Floor field model	A very simple evacuation scenario	Coupling sensitivity parameters to the dynamic and static floor fields, respectively	Herding behavior; Familiarity with geometry
2	Cellular automaton model for evacuation process with obstacles [42]	Floor field model	A classroom with a single door and double doors	A random number decides the conflict between two pedestrians	Obstacles distribution; Panic variable
3	Paired behavior effect on pedestrian evacuation dynamics [43]	Floor field model	A room with two exits	Pairs as a whole; Six movement directions without considering diagonally backward	Side-by-side in paired; Front-behind in paired; Mmixed paired; Paired ratio
4	A dynamic cellular automaton model for evacuation process with obstacles [44]	Floor field model	A restaurant and a classroom with door's position changed	Cells weight affected by crowd distribution; An experiential value used to avoid a deterministic model	Crowd distribution; Doors position; Doors width
5	Evacuation of pedestrians from a single room by using snowdrift game theories [45]	Lattice gas model	A square room	Game theory (pedestrian can move to both empty and occupied sites)	Pedestrian density; Cost-to-benefit ratio
6	A study of pedestrian group behaviors in crowd evacuation based on an extended floor field cellular automaton model [35]	Floor field model	A university building a hall and an adjacent passage leading to an emergency exit of a university building	Leader-follower behavior rule	Group size; evacuation Decisions dependency strengths; Crowd density
7	Modeling pedestrian flow accounting for collision avoidance during evacuation [46]	Lattice gas model	A classroom with two exits	Collision avoidance behavior	Avoidance behaviors; Competition behaviors; Layouts
7	An extended cost potential field cellular automaton model for pedestrian evacuation considering the restriction of visual field [47]	Floor field model	A room with one exit	Visibility function introduced to describe visual effect; Cost potential field considering behavior variation	Psychology tension; Visual radius; Pedestrian density
8	Relationship between behavior aggressiveness and pedestrian dynamics using behavior-based cellular automata model [48]	Floor field model	A long corridor with one exit	Degree of aggressiveness introduced to determine the balance selection between keeping shorter paths and avoiding congestion	Shortest path tendency; Detours tendency; Aggressiveness
9	Influence of human-obstacle interaction on evacuation from classrooms [49]	Floor field model	Traditional classrooms with three types of aisle layouts; active learning classroom with four layouts	Two types of human-obstacle interactions: Passing-over-obstacles and pushing-obstacles.	Exits and aisles; Local density; Personality traits
10	An extended model for crowd evacuation considering pedestrian panic in artificial attack [50]	Floor field model	A room with one exit	Three panic behaviors in artificial attack	Panic control coefficient; Pedestrian initiative and passivity; Initial position of the attacker
11	Simulation of a pediatric hospital in evacuation considering groups [51]	Floor field model	The first floor of outpatient department of the pediatric hospital	Local density field; Overlapping and speeding up rule; Priority order; Cohesive distance within groups	Pedestrian type
12	Integration of cumulative prospect theory in cellular automata model for building evacuation [52]	Floor field model	A L-shaped corridor with two exits; a square room with one exit	Cumulative prospect theory for decision-making	Herding; Inertia
13	Real-time building evacuation modeling with an improved cellular automata method and corresponding IoT system implementation [53]	Floor field model	The west side building of the Beijing Capital Airport Emergency Center	Introducing real-time fire point into a potential energy field	Exit width; Number of escaped; Fire point
14	Simulation of adult-child matching behavior: A case study of primary school in Linyi city, China [54]	Floor field model	Simplified middle school layout	Different field floors before and after matching for both adults and children	Matching strategies
15	An evacuation model considering the phototactic behavior of panic pedestrians under a limited visual field [55]	Floor field model	A room with one exit	Emotional contagion mechanism; guidance field, orienteering field, herding field, and phototactic field	Panic; Emotional thresholds; Information transfer; Visual field distance; Light sources position

**Fig. 6.** Schematic diagram of the social force model.

applicability in diverse and complex evacuation scenarios, enabling it to comprehensively simulate the intensity range and dynamic changes of different disasters, as well as the dynamic change processes of the population.

Li et al. [86] simulated evacuation panic during the 2013 Ya'an earthquake in China and analyzed the maximum crowd pressure at different speeds. Bernardini et al. [87] developed a joint model based on shallow water fluid dynamics, demonstrating the impact of fixed obstacles on pedestrian movement direction during flood disasters. Lu et al. [88] used a university campus as a case study to investigate the impact of falling objects on pedestrian tripping behavior during earthquakes. Zhou et al. [89] modified the social force model based on their research on the quantitative panic coefficient and initial expected speed calibration, which more accurately simulated the crowd evacuation process following an earthquake. Zhang et al. [90] designed a social force model that considers the subjective initiative of pedestrians to adapt to complex subway station fire environments with multiple exits and floors. Zhong et al. [91] used the central urban area of an earthquake-prone city in China as a case study to conduct large-scale regional evacuation while considering the spatial heterogeneity of the age structure and helping behavior. Shao et al. [92] established an evacuation model for simulating chemical disasters, developing a pedestrian deployment strategy based on social relationships for large-scale evacuation.

In crowd evacuation scenarios, the social force model effectively represents the continuous movement of individuals, accounting for both self-influence and environmental interactions. Despite its complexity, advancements in computational power are broadening the model's application in evacuation studies, enhancing its utility in capturing the intricacies of crowd dynamics. The social force model can be adjusted and expanded according to actual situations to adapt to different behavioral patterns and environmental conditions. This flexibility enables the social force model to address various simulation needs effectively. The improvements and optimizations made by scholars have enabled the simulation of real disaster events in complex and dynamic large-scale environments.

3) Agent-based Model

Agent-based modeling (ABM) offers a unique approach to crowd simulation, distinguishing itself from rule-based, grid-centric cellular automata [36] and the continuous, force-focused social force model

[67]. Emerging as a key method in artificial intelligence research for disaster scenarios, ABM's strength lies in its detailed representation of individual agents, each with distinct behaviors and capable of complex interactions [93], as illustrated in Fig. 7. This individual-focused framework allows for a more nuanced simulation of crowd dynamics, effectively capturing both individual decision-making and collective behavior, which is less emphasized in cellular automata and social force models. For example, Sharma et al. [94] developed a simulation approach to modeling realistic human behavior during evacuation drills using each intelligent agent's characteristics of independence, collective ability, cooperativeness, and learning. Lim et al. [95] defined different behavior rules and decision-making processes for each agent in six categories: elderly, caregivers, nurses, physical therapists, social workers, and visitors, allowing for behavioral interactions among different types of personnel during the evacuation process. In emergency evacuations, the ABM's ability to replicate real-world human interactions makes it an invaluable tool for understanding and managing complex evacuation scenarios.

Evacuation simulations utilizing ABMs have significantly evolved. Bonabeau [96] highlighted the efficacy of ABM in capturing real-world system dynamics, especially in depicting diverse behavioral patterns such as panic, offering more flexibility than other models despite higher computational demands. Pan et al. [97] developed a multi-agent system to simulate various social behaviors during evacuations, including herding and competitive actions. Chen et al. [98] focused on crowd evacuation in panic situations, examining the effects of panic on pedestrian behavior and evacuation efficiency. Hassanpour and Rassafi [99] utilized an intelligent agent model for subway station evacuation under various conditions, assessing maximum evacuation flow rates. Gobbin et al. [100] analyzed how factors such as walking speed and exit allocation affect evacuation time. Bao and Huo [101] introduced an advanced model considering rotational behavior, exploring the interactions between evacuees and their environment. Cotfas et al. [102] applied ABM to large event hall evacuations, incorporating diverse agent characteristics and proposing an "adapted cone exit" strategy for efficient exit guidance. Wang et al. [103] devised a multi-agent system based on decision theory, integrated into the social force model of AnyLogic software to simulate emergency evacuation scenarios, highlighting behaviors such as cooperation, obstacle avoidance, and exit competition. These studies collectively enhance the understanding and application of ABM in various evacuation contexts, highlighting its versatility in modeling complex human behaviors.

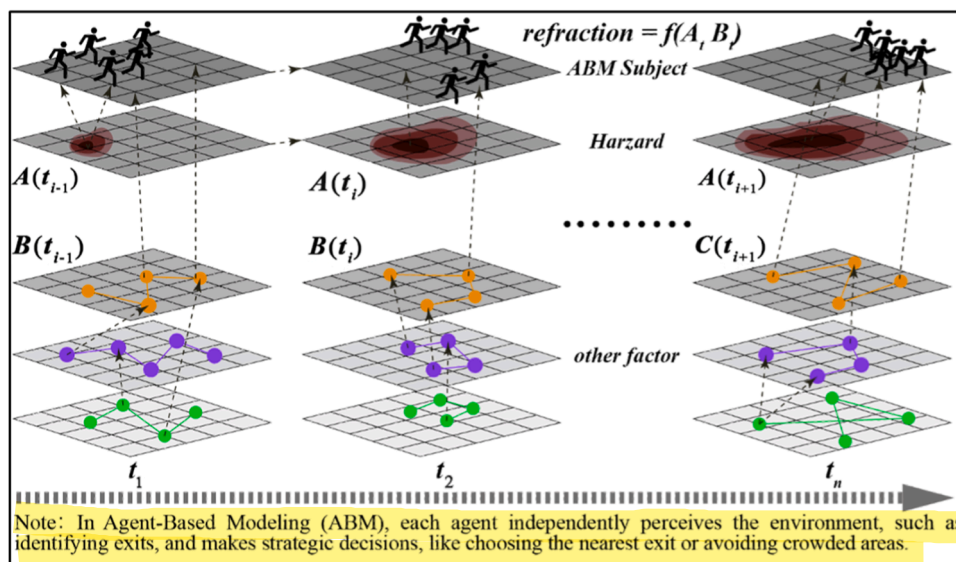


Fig. 7. Schematic diagram of the agent-based model.

Simultaneously, dynamic environments are increasingly being incorporated into agent-based models by researchers to simulate complex behaviors in response to changing scenarios. **Agent-based models are suitable for exploring human behavior and rapid low-level environmental changes via available high-resolution data.** Mas et al. [104] applied this model to evaluate tsunami risk and evacuation scenarios in Indonesia, Thailand, Peru, and Japan, considering the preferences for the fastest and lowest tsunami risk options. Yang et al. [105] employed agent-based modeling to simulate the evacuation of deep foundation pits and studied the behavior of avoiding falling objects during collapse disasters. Wang et al. [106] utilized mutual assistance mechanisms to analyze the flood evacuation process in vulnerable communities. Shirvani et al. [107] introduced an agent-based simulator to dynamically model flood-people interactions. Grajdura et al. [108] developed an ABM for the 2018 Northern California wildfire evacuation, employing post-disaster surveys and **decision tree methods** to authentically capture evacuee behavior. Chapuis et al. [109] explored a model specifically designed to evaluate spatiotemporal large-scale evacuation strategies applied in inland areas, considering mixed traffic choices and applying it to flood prone-areas.

As described in the above research, the agent-based model is beneficial for major disaster mitigation and evacuation planning by describing the personalities of evacuees and allowing for the observation of emergency behavior in the dynamics of agent interactions. The ABM can more flexibly model changes in space and time, which enables the model to better simulate the spatiotemporal interaction in large-scale evacuation. These developments signify a sophisticated application of ABM in capturing the nuanced interplay between agents and dynamic environments in emergency scenarios.

4) Combination of evacuation models

Ongoing research strives to enhance the accuracy and realism of simulations. Recognizing that **a single model may not capture the full complexity of real-world evacuation** [46,85,103], recent efforts have concentrated on integrating various model types. This **amalgamation** of different evacuation models yields a more holistic and precise tool, better suited for responding to emergencies and disasters. This section aims to explore and summarize these combined modeling approaches, as shown in Table 4, highlighting their benefits and addressing the challenges they may face in practical applications.

Integrating multiple evacuation models has significantly advanced simulation research by capitalizing on the unique strengths of different approaches. For example, **social force models excel in simulating interpersonal dynamics** [68], while **agent-based models adeptly capture individual decision-making** [93]. This synergy enhances the comprehensiveness and precision of evacuation tools, offering a more nuanced understanding of collective evacuations and improving preparedness for emergencies. However, challenges persist, including the **complexity of model integration and increased computational demands**. Future research should focus on refining integration methods and addressing these challenges to elevate the accuracy and practicality of evacuation simulations, thereby supporting more effective emergency decision-making and planning.

3.1.2. Macroscopic models

The macroscopic evacuation models aim to simulate the overall behavior and flow of large-scale crowds without involving the details of individual behavior. These models are usually based on physical principles, fluid dynamics theory, or network structures to predict the evacuation process and results of crowds. They are particularly suitable for simulating large-scale evacuations and capturing the overall crowd dynamics. **Pedestrian flow continuum evacuation models** and **network flow evacuation models** are two widely used models. Macroscopic models under major disaster scenarios need to consider the impact of the disaster itself on the environment and people, such as the impact of

Table 4

Studies that integrate various evacuation modeling techniques.

No	Ref.	Models	Innovations	Advantages
1	Simulation of evacuation processes using a multi-grid model for pedestrian dynamics [110]	LG; SF	A finer lattice; Each pedestrian occupies multiple grids; Quantifying forces	High calculation speed; Simulation of pedestrian interaction
2	A mobile lattice gas model for simulating pedestrian evacuation [111]	LG; SF	Mobile lattice positions	Ability of accurately computing moving distance and evacuation time; Simulation of pedestrian interaction
3	Study on crowd evacuation behaviors based on multi-agent and cellular automata technology [112]	ABM; CA	Treating each cellular as an agent; Autonomous agents with complex decision-making and judgment abilities	Better reflects real behaviors of the crowd during evacuation process
4	Cellular automaton model for social forces interaction in building evacuation for sustainable society [113]	CA; SF	Clustering attraction	Reducing time by adjusting the social behaviors of evacuees
5	Variable guiding strategies in multi-exits evacuation: Pursuing balanced pedestrian densities [114]	FF; SF	Adjustable guiding attractions	Suppress unbalanced pedestrian distribution
6	An extended model describing pedestrian evacuation considering pedestrian crowding and stampede behavior [59]	FF; SF	Crowding force and friction	Revealed mechanism and influencing factors of stampede accidents
7	Improved Social Force Model Based on Enhancing Psych behavioral Heterogeneity [115]	ABM; SF	Using SFM to analyze changed data perceived by agent to obtain new motion vectors; planning new motion paths in turn	More authentic; Improved simulation accuracy

Notes: LG—Lattice gas model; SF—Social force model; CA—Cellular automata model; ABM—Agent-based model.

building damage and obstacle distribution on road network capacity under earthquakes and the impact of smoke on pedestrian flow speed under fire scenarios.

The macroscopic evacuation models are designed to simulate the overall behavior and flow of large-scale crowds without involving the details of individual behavior. These models typically rely on physical principles, fluid dynamics theory, or network structures to predict the evacuation process and outcomes of crowds. They are particularly suitable for simulating large-scale evacuations and capturing overall crowd dynamics. Pedestrian flow continuum evacuation models and network flow evacuation models are the two most widely used models. Macroscopic models for major disaster scenarios must consider the disaster's impact on the environment and people, such as the effects of building damage and obstacle distribution on road network capacity during earthquakes and the influence of smoke on pedestrian flow speed during fires.

1) Pedestrian Flow Continuum Model

Pedestrian flow continuum evacuation models treat crowds as fluid-like entities, describing their collective behavior based on fluid dynamics principles. These models consider the macroscopic flow characteristics of crowds, such as density, velocity, and pressure, without explicitly representing individual pedestrians. This type of model does not provide a network abstraction of the evacuation environment. Notably, it completely treats humans as fluid particles, with human flow determined by mass conservation equations, initial conditions, and boundary conditions. Pedestrian flow research was developed as part of traffic flow research, with scholars using fluid dynamics models from traffic flow studies [116]. Dedicated to exploring indicators such as population density and speed, the relationship between these indicators is termed the fundamental diagram [117]. In 1971, Henderson [118] first applied the macroscopic theory of fluid mechanics to pedestrian dynamics. Hughes [119] defined crowds as "thinking fluids" and introduced a first-order model based on fluid dynamics equations, coupling the standard mass conservation equation with the Eikonal equation based on density distribution within the domain to describe movement direction, thus establishing a theoretical framework for a continuous pedestrian flow model. Narain et al. [120] proposed a macroscopic continuum model and made it possible to model dense crowds composed of up to a hundred thousand pedestrians. Pedestrian flow models can be used for overall research on large-scale evacuations in major disasters. First-order models, supplemented by simplified equilibrium velocity-density relationships, have been criticized [121] for their inability to reproduce complex crowd phenomena such as congestion. Twarogowska et al. [122] employed a second-order macroscopic model of crowd dynamics, reproducing phenomena such as stop-and-go waves and clogging at bottlenecks. Considering parameters such as the exit velocity, walking speed, exit velocity coefficient, and combined flow ratio in the continuous flow equation, as well as uneven pedestrian distribution, Wu et al. [123] studied Taipei 101, calculating total and phased evacuation times and the number of people on each floor over time. Maity and Sundar [124] combined the second-order continuum macroscopic description of the crowd evacuation model with the fluid dynamics model of flood inundation described by the two-dimensional depth-averaged shallow water equation (SWE), considering bidirectional dynamic coupling, to study collective crowd behavior in sudden flood evacuation scenarios. The time history of evacuees' risk states, evacuation patterns under different flow conditions, and the effects of stress-induced behavior are subsequently revealed. Jiang et al. [125] proposed a viscous continuum pedestrian flow model incorporating the effects of smoke to simulate pedestrian evacuation in smoke-filled facilities during fires, considering viscosity to express pedestrian resistance to speed changes. Zhu et al. [126] used the population parameters of the social force model as inputs for the macroscopic model, combining the advantages of both to simulate the 2015 Mecca Hajj stampede.

2) Network Flow Model

The network flow model, derived from network flow theory [127], is designed on a network of human flows with spatial constraints and specific objectives, being a widely applied and well-established macroscopic evacuation model. It is frequently employed to develop optimal evacuation plans, with the most common application being the planning and allocation of personnel evacuation routes. To model evacuation dynamics, it relies on an optimization model to find the optimal evacuation dynamics between locations with complex terrains. The network flow model abstracts interconnected buildings into a static network and extends it into a spatiotemporal network to capture evacuation dynamics. Finally, an objective function is established on the dynamic network, with conditions such as bottlenecks and bidirectional flows selected as constraints [128]. Various methods have been studied to effectively solve practical problems through different objective functions and constraints. Fracis [129] established a "uniformity principle" without considering road network capacity, significantly analyzing the

shortest evacuation time and corresponding route allocation. Kamiyama et al. [130] proposed a fastest flow problem in a network composed of directed graphs with capacity and transit times on arcs, presenting a time algorithm for solving the problem with uniform arc capacity. Shin and Moon [131] adopted collapsible nodes and arcs. Given that placing facilities in arcs would reduce the capacity, Nath et al. [132] introduced precise algorithms for solving single-facility situations and heuristic algorithms for solving multi-facility situations. Many studies have focused on diversion, with one of its main tasks being to identify alternative routes that can divert traffic congestion by best utilizing existing infrastructure capacity. Zheng et al. [133] established an evacuation crowd diversion model, formulating reasonable rules for pedestrian flow allocation at road intersections. Cui et al. [134] developed a new diversion routing model based on the minimum cost network flow model, considering the characteristics of traffic diversion during evacuation. A prominent feature of the new model is the introduction of intersection control costs of diversion operations into the route optimization process, making it more suitable than conventional models for determining actual diversion routes in evacuations. Scholars have improved and optimized network flow models for various research objectives and application scenarios. Chen and Lin [135] proposed a dynamic network-based optimization algorithm to provide optimized evacuation plans and clearance times for residents in flood retention basins, considering simulation results, sensitivity, and scenario analysis. Lu et al. [136] proposed a game-theory-based approach to analyze network vulnerability, considering travel costs and flooding risks on evacuation routing under a hurricane. Zhang et al. [137] established a population road risk quantitative assessment system focused on major earthquakes, considering eight influencing factors, including road width, node degree, safety interval, road resistance coefficient, building threat, pedestrian backflow, illegal parking of vehicles, and traffic flow. Wang et al. [138] proposed a new network-flow-based migration strategy model, comprehensively considering factors such as path classification, resettlement capacity constraints, population units, and path congestion, in response to the optimization problem of flood avoidance and evacuation paths during flood disasters. Jin et al. [139] used a network flow model and developed a minimum cost maximum flow method to effectively utilize existing underground spaces for large-scale pedestrian evacuation in disaster situations. Shin and Moon [131] utilized the robust evacuation planning function of dynamic network flow to analyze whether the optimal evacuation path for pedestrians will pass through the collapsed area of a building. If the timeframe is sufficient, the model allows pedestrians to detour; if the time is insufficient, the model assumes the risk and provides a route including the collapsed area.

Overall, the pedestrian flow model emphasizes the overall behavioral characteristics and movement patterns of the crowd, while the network flow model emphasizes the spatial structure and path planning during major disasters. The primary advantage of these macroscopic models is their ability to capture the global behavior of large crowds, rendering them well-suited for analyzing scenarios involving large-scale venues such as stadiums, transportation hubs, or urban areas. These models are particularly valuable for emergency management and urban planning because they can facilitate the design of efficient evacuation strategies, the identification of potential bottlenecks, and the evaluation of the effectiveness of infrastructure improvements.

3.2. Enhanced scenario simulation tool

The conduct of evacuation drills involving numerous personnel for unforeseen events poses significant challenges. Such drills often fail to reflect the psychological state of individuals in real emergencies, are resource-intensive, and cannot cover all potential scenarios [140]. In contrast, evacuation models, combined with computer simulation technology, provide a potential solution. These computer-based simulations provide a wealth of visual, repeatable, and reusable data, as

demonstrated by the papers in Table 5. They visually simulate evacuation processes and provide intuitive criteria for evaluation. Importantly, these simulations reveal unpredictable human behaviors during emergencies. Over 20 evacuation models and corresponding software have been developed by international scholars, with several key examples presented in Table 5.

Evacuation software such as Pathfinder, BuildingEXODUS, and SIMULEX are widely utilized in research, yet these tools possess certain limitations [168]. Pathfinder software has been found to have impractical parameter settings, whereas BuildingEXODUS uses default values that may not be suitable for all scenarios, necessitating careful verification. Additional tools are needed for SIMULEX to perform comprehensive fire and smoke evacuation analysis. Despite the noted challenges, these evacuation software tools remain valuable for the analysis of evacuation strategies, the optimization of building designs, and the study of emergency human behavior. The selection of software is dependent on the specific requirements of the application.

3.3. Crisis simulation and effectiveness evaluation strategies

The need for effective emergency evacuation strategies in response to the increasing impact of three major disaster types—fires, earthquakes, and floods—in densely populated urban areas is critically examined in this paper. The significance of these disasters has been underscored by Fig. 8, which demonstrates their frequency and proportion relative to all disaster types [1]. These three typical disasters exhibit varying characteristics in terms of impact scope, intensity, and dynamic changes, thereby guiding the development of targeted evacuation strategies and simulations for these specific scenarios.

3.3.1. Fire scenarios

Emergency evacuation research has focused on fire scenarios, where predicting human behavior is challenging due to its unpredictability [169]. Simplified computer models are being developed by experts worldwide to improve evacuation planning, as demonstrated in Fig. 9. [170]. The impact range of a fire depends on its size and spread speed, which can spread from a single building to a large area. The spread of a fire and its smoke is characterized by rapidity and a high degree of urgency, necessitating immediate evacuation action.

1) Smoke-Intensive Scenarios in Fire Emergency Evacuation Simulations

Understanding smoke dynamics is considered crucial within the

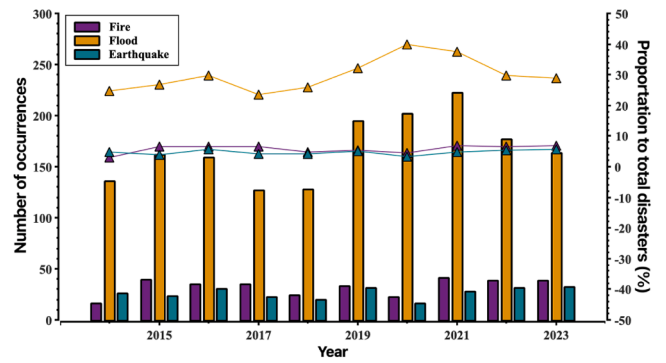


Fig. 8. Global frequency of fires, floods, and earthquakes included in EM-DAT, 2004–2023.

realm of fire research, as smoke often leads to significant casualties. Rapid changes in visibility, temperature, and carbon monoxide levels within a burning building critically impact evacuation efforts, ultimately resulting in substantial economic loss and human fatalities [169]. These factors must be accounted for in effective emergency evacuation strategies, as they significantly influence group evacuation timing and route selection.

Various models have been developed to address different aspects of smoke and fire dynamics. Yang et al. [171] utilized a cellular automaton model to simulate crowd evacuation, incorporating hazard levels and field of vision. Yan and Tan [172] focused on visibility reduction due to smoke, employing a "follow the leader" approach in their cellular automaton framework. Peng and Chou [173] explored smoke effects in multi-level buildings, particularly congestion at critical points like T-junctions. Zhang et al. [174] utilized the Monte Carlo method in their evacuation model, which was validated against fire drill data. Guo et al. [175] developed a fire-pedestrian interaction (FPI) model, highlighting the influence of temperature on evacuation routes. Mu et al. [176] conducted dual-tunnel fire evacuation experiments, with simulations demonstrating good agreement with real data. Zheng et al. [177] introduced an extended ground field model to assess the impact of fire and smoke on evacuation efficiency. Kodur et al. [178] analyzed occupant evacuation in a 32-story building, finding that lower floor fires significantly affect evacuation times. Wang et al. [179] employed BuildingEXODUS for evacuation simulations, correlating fire location to optimal exit strategies. Deng et al. [6] performed simulations on pre-fabricated construction sites, focusing on visibility, carbon monoxide levels, and temperature. Liu et al. [148] integrated BIM and FDS for

Table 5
Summary of commonly used evacuation software.

Category	Software			
	Pathfinder	AnyLogic	BuildingEXODUS	SIMULEX
Underlying model	ABM	ABM; SD; DE; SF (Pedestrian library)	CA	ABM
Function and Process	Scene construction, pedestrian setting, simulation, 2D and 3D dynamic display, indicator chart output	Scene construction, pedestrian settings, dynamic parameter settings, simulation, 2D and 3D dynamic display, dynamic chart statistics	Scene construction, pedestrian setting, simulation, and result output	Scene construction, pedestrian settings, simulation, 2D and 3D dynamic display, indicator charts and outputs
Extensibility	Support the import of CAD and FDS files; Non open source	Support external data sources such as GIS and CAD; Connect to the database; Support custom model development using Java; Non open source	Support the import of CAD scene files; Non open source	Support the import of CAD scene files; Non open source
Applicable scenarios	Fire evacuation; Building evacuation; Traffic evacuation	Suitable for various domains such as transportation, logistics, manufacturing, healthcare, and supply chain for simulating complex systems and optimization analysis	Suitable for simulating evacuations in buildings and enclosed spaces for analyzing evacuation strategies, pedestrian flow, and fire scenarios	Suitable for large public buildings and facilities for simulating evacuations, analyzing evacuation strategies, building layouts, and fire scenarios
Papers	[141–148]	[149–157]	[158–163]	[164–167]

Notes: CA—Cellular automata model; SF—Social force model; ABM—Agent-based model; SD—System dynamics; DE—Discrete event simulation.

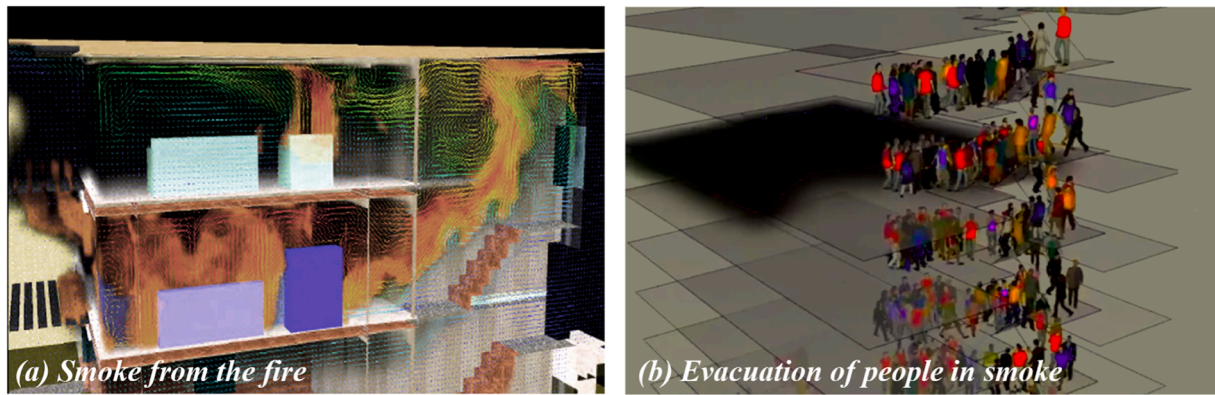


Fig. 9. Simulation of pedestrian evacuation in fire scenarios. (a) Smoke from a building fire; (b) Evacuation of people in smoke.

underground tunnel evacuation studies. However, a gap remains in simulations for university building complexes, where occupant density and safety awareness are critical factors. Gao et al. [180] addressed this by applying BIM technology and Pathfinder software to optimize evacuation plans for such complexes.

2) Strategic Planning of Evacuation Paths in Fire Emergencies

In fire emergencies, panic and herding behavior, especially among those unfamiliar with buildings, often hamper effective evacuation. Dynamic evacuation path planning, tailored to fire progression, can significantly enhance evacuation efficiency and individual safety. The evolution of fire evacuation path planning has demonstrated significant advancements, progressing from foundational behavioral studies to sophisticated algorithms. Initially, research in this area focused on understanding human behavior during fire emergencies, particularly how individuals and groups respond to fires and make decisions about evacuation [130,132]. This foundational work was crucial for developing initial guidelines and strategies for effective evacuation planning [143].

With technological advancements, the focus shifted to developing more sophisticated methods for evacuation planning. This transition entailed the utilization of computer simulations and models to predict fire scenarios and identify optimal evacuation routes. Various factors, such as the layout of the building, the location of exits, the intensity and spread of the fire, and human behavior during emergencies, are considered in these models [133,134]. In recent years, the incorporation of big data, artificial intelligence, and machine learning has further transformed this field. These technologies enable more dynamic and adaptive evacuation planning, incorporating real-time data and potentially changing scenarios during a fire. Targeted information's impact

was studied by Tucker et al. [181], and innovative algorithms were developed by Zhou et al. [182] and Yen et al. [183]. Most recently, Deng et al. [6] introduced real-time fire spread analysis for evacuation routing. This progression demonstrates a shift from basic models to data-driven approaches, although challenges in personnel parameterization and algorithm refinement remain. The use of these advanced algorithms may result in more effective and efficient evacuation strategies, potentially leading to increased life-saving outcomes during fire emergencies.

3.3.2. Earthquake scenarios

The study of earthquake evacuation is crucial because of its unique challenges. Unlike other disasters, earthquakes strike suddenly, causing immediate structural damage and obstructing evacuation routes [3], as illustrated in Fig. 10. (a). Earthquakes, always resulting in building collapse, can cover vast areas in a short period and have extremely high urgency, requiring immediate evacuation actions. Consequently, targeted research to develop tailored evacuation strategies for earthquakes is essential, given their unpredictability and the significant risks they pose.

1) Seismic evacuation simulation with earthquake-induced pathway disruptions

Earthquakes profoundly disrupt urban environments, with road blockages caused by damage to infrastructure severely hindering evacuation efforts [184,185]. Such seismic-induced congestions are identified as key factors complicating the evacuation process, emphasizing the critical need for efficient seismic evacuation paths to facilitate safe relocation and disaster relief.

Seismic evacuation modeling has undergone significant evolution

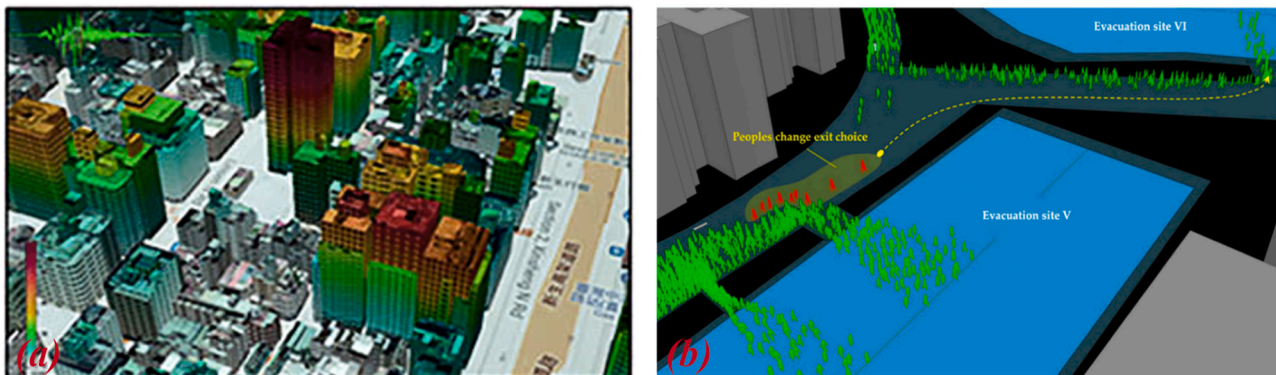


Fig. 10. Simulation of pedestrian evacuation in earthquake scenarios. (a) Simulation of building damage in earthquake scenarios; (b) Simulation of pedestrian evacuation in fire scenarios.

over the past several decades, marked by key contributions at each stage. By 2012, Mas et al. [186] developed a more nuanced model via NetLogo simulation software, integrating GIS and multi-agent systems, specifically for the 2011 Great East Japan Earthquake. The mid-2010s saw further advancements with researchers like Bernardini et al., who analyzed evacuation behavior using intelligent agent models [187], and Xiao et al., who combined building finite element models with social force evacuation models [188]. Ertugay et al. [185] developed methods to estimate the probability of road closures during earthquakes. The late 2010s focused on infrastructure, with Aguado et al. [189] predicting building damage and road blockage scenarios and Santarelli et al. [190] assessing post-earthquake road network capacities. The early 2020s saw a shift toward more detailed behavioral analyses and community-focused solutions. Xiao et al. [191] and Lu et al. [88] delved into the influence of building dynamics on evacuation, whereas Feng et al. [192] utilized virtual reality to study individual decision-making during earthquakes. Most recently, Yang et al. [193] introduced the concept of "virtual evacuation walls," and El-Maissi et al. [194], along with Amini et al. [184], developed models considering the broader impact of earthquakes on evacuation processes. Liu et al. [3] further expanded the field by analyzing the impact of community building debris on evacuation and proposing optimized designs for emergency spaces. This chronological narrative underscores the field's progression from basic theoretical models to comprehensive, context-specific simulations that address both infrastructure and human behavior in seismic evacuations.

Earthquakes present a complex challenge for evacuation due to their sudden onset and potential to trigger secondary hazards in urban environments. Improving the accuracy of pedestrian dynamics simulations in evacuation models has been a key focus of recent research, addressing the complexities of human decision-making and behavior during these emergencies. This work is crucial for enhancing evacuation efficiency and reducing casualties in earthquake scenarios.

2) Optimizing Evacuation Pathways with Earthquake Shelter Space Considerations

In seismic-prone areas, the optimization of evacuation routes with an emphasis on earthquake shelter spaces is considered crucial. Rapid and efficient evacuation to designated safe zones, such as shelters and open spaces, is illustrated in Fig. 10. (b), is essential during sudden earthquakes. Pre-planning safe spaces and evacuation paths can significantly enhance the effectiveness of disaster response and emergency management, thereby ensuring the safety of affected populations during earthquake emergencies.

Seismic evacuation research has evolved from basic shelter location choices to intricate, targeted evacuation strategies. Initially, Alcada-Almeida et al. [195] highlighted the strategic placement of shelters and rescue facilities. Building upon this, Bayram et al. [196] introduced quantitative models designed to minimize evacuation time by considering the capacity and location of shelters, particularly in urban contexts like Istanbul. Significant advancements were observed in the late 2010s, with Esposito Amideo et al. [197] and Yamamoto and Shimura [198] incorporating GIS and multi-objective optimization into evacuation planning. Their work entailed the allocation of emergency routes and shelters based on varied scenarios, utilizing advanced computational methods for more effective planning. In the early 2020s, the focus shifted to more specific environments and refined methodologies. Zhang et al. [199] analyzed evacuation requirements in campus settings, optimizing space and routes for efficiency and safety. Concurrently, Wei et al. [200] tackled the challenges posed by limited shelter capacities and population distribution constraints, developing a model to minimize evacuation distance. Keykhaei et al. [201] employed multi-agent simulation to establish a situational awareness system for identifying, inferring, and extracting different levels of catastrophic earthquakes before and after an event, thus facilitating rapid decision-making and

timely evacuation responses. Finally, Liu et al. [3] proposed optimized designs for emergency spaces in high-density communities, emphasizing time-efficient evacuation organization. This developmental arc in seismic evacuation research demonstrates a trend toward integrating sophisticated technology and scenario-based planning, with the aim of achieving more precise and effective evacuation strategies in diverse urban and community settings.

3.3.3. Flood scenarios

In the context of flood disasters, the primary emphasis is placed on developing effective evacuation strategies. While modern advancements have led to precise flood prediction models [124], the critical challenge of safely evacuating large populations through complex networks persists, as illustrated in Fig. 11. (a). Although the urgency of floods is relatively low, considering the rapidly changing and unpredictable nature of rising water levels, as depicted in Fig. 11. (b) [202], may render buildings and routes inaccessible over time. Consequently, effective flood evacuation strategies must be dynamic and adaptable, specifically tailored to address these distinct and time-sensitive challenges posed by flood scenarios.

1) Advanced Simulation of Emergency Evacuations in Response to Flood Inundation Dynamics

The evolution of emergency flood evacuation models has advanced from basic behavioral simulations to sophisticated, real-world scenario integrations. Initially, Simonovic and Ahmad [203] set the foundation with a model focusing on human behavior during floods via system dynamics. This early work paved the way for more detailed studies on decision-making in flood scenarios. Guo and Huang [204] and Bernardini et al. [87] investigated how people choose exits and the impact of flood conditions on pedestrian speed. The mid-2010s witnessed an expansion of research on the dynamics of information propagation and interactions between individuals and their environment during floods. Du et al. [205] and Li et al. [206] utilized multi-agent systems and cellular automata to simulate these complex interactions, offering deeper insights into crowd behavior during floods. In the late 2010s, there was a specialized focus on vulnerable groups, with studies like those of Lee et al. [207] examining the evacuation speed of the elderly and Zheng et al. [208] exploring the unique challenges faced in underground environments. The 2020s marked a significant shift toward integrating real-world behaviors into evacuation models. Nakanishi et al. [209] employed intelligent agents and geometric networks to more accurately mirror resident behaviors, whereas Huang et al. [210] developed an advanced flood inundation analysis model using cellular automata, focusing on the effects of flood spread on pedestrian movement. Most recently, Maity and Sundar [124] proposed a model that innovatively combines crowd evacuation dynamics with the fluid dynamics of flood inundation, highlighting bi-directional dynamic coupling. These latest advancements reflect the ongoing efforts to create more comprehensive, realistic models that can effectively inform urban flood prevention strategies and emergency evacuation planning, with ample room for future enhancements and refinements.

2) Strategic evacuation planning in response to dynamic flood inundation

Over the past few decades, flood evacuation strategies have evolved from basic logistical planning to sophisticated, adaptive approaches in response to the increasing complexities of flood disasters. Southworth and Chin [211], in the 1980s, laid the groundwork with fundamental strategies focusing on effective routes and site selection for evacuation. This approach was furthered in the 1990s by Dunn and Newton [212], who introduced a systematic analysis using GIS to optimize evacuation paths based on road capacity. Since the 2000s and 2010s, the focus has shifted toward enhancing efficiency and practicality, as evidenced by

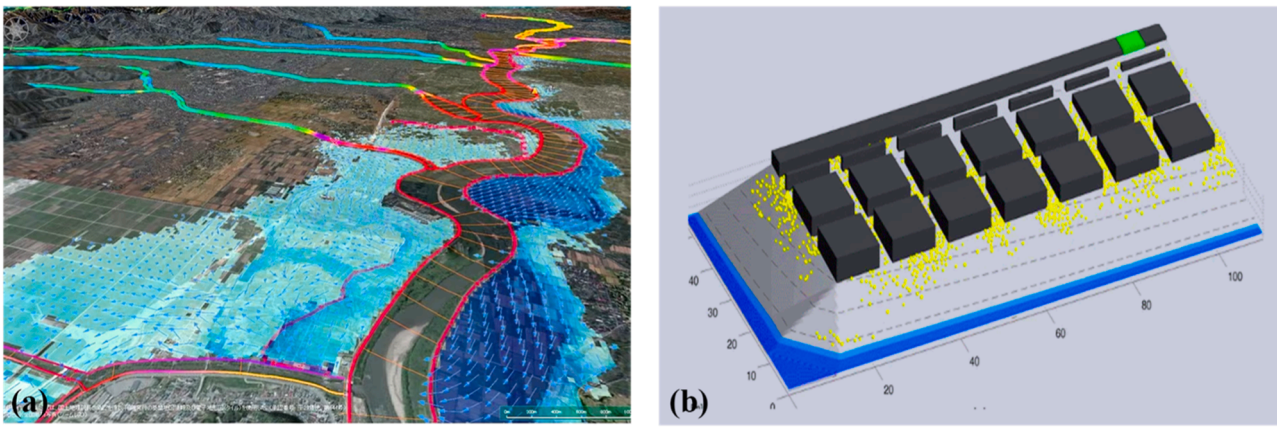


Fig. 11. Simulation of pedestrian evacuation in flooding scenarios. (a) Flood inundation simulation; (b) Simulation of water level changes in flood scenarios.

the shortest path transfer plan [213] and models that consider factors such as road capacity and safety for efficient evacuation routes [214]. The integration of advanced technologies for strategic planning gained prominence in the 2010s and 2020s, exemplified by network analysis [215] and the development of practical evacuation maps [216]. Most recently, the 2020s have seen a shift toward adaptive models such as those developed by Li et al. [4] and Chelariu et al. [217], which address the dynamic and unpredictable nature of floods, focusing on flexible and resilient strategies. This trajectory reflects an increasing emphasis on not only efficiency but also adaptability and robustness in evacuation strategies to address the challenges of modern flood scenarios.

3.4. Emergency evacuation simulation and strategy summary for different disasters

Aiming at different types of major emergency disaster events, assessing their impact scope and intensity, as well as monitoring and simulating the behavioral changes of affected people, is the key to formulating effective emergency evacuation strategies. Each disaster has unique characteristics and different levels of urgency; therefore, it is necessary to develop corresponding evacuation models to obtain targeted strategies based on the characteristics and impact of the disaster. Through scientific assessment and simulation, response measures can be improved, resulting in improved efficiency of emergency responses and maximized protection of personnel safety. In the following section, the impact scope, emergency degree, and corresponding evacuation simulation methods and evacuation strategies for three typical major disasters are discussed.

Fires, as common and serious destructive sudden disasters, generally affect small areas and cause a high degree of emergency. The impact of a fire can range from individual buildings to regional fires and widespread wildfires. Smaller fires may produce smoke and flames locally, whereas large-scale fires can cause buildings to collapse, spread rapidly, and emit substantial heat and harmful gases, threatening personnel safety. Predicting the development and specific conditions of a fire is often challenging. Regardless of size, fires tend to spread rapidly, necessitating urgent fire evacuation procedures. Fire evacuation modeling can be based on the building structure [218] and employs fire simulation software or the model itself to simulate factors such as fire source location [219] and smoke diffusion [220]. Fire evacuation is often modeled from a microscopic behavioral perspective [90]. Microscopic models, such as cellular automata, are combined to simulate the impact of fire dynamic changes, including smoke diffusion and temperature fluctuations, on crowd behaviors [60]. Smoke reduces visibility, making breathing and escape difficult. The model should account for physical conditions [221], including walking speed, escape skills and mobility. Owing to the urgency of fire evacuation, pedestrians must be evacuated

to minimize exposure to fire smoke based on the shortest evacuation time possible.

Earthquakes are characterized by extreme severity, irrespective of their impact area or level of urgency. Strong earthquakes possess significant destructive power and are often accompanied by secondary disasters. The urgency of earthquakes is utmost high, as damage and casualties can expand rapidly after an earthquake occurs. Situations such as collapsed buildings, trapped people, blocked roads and power outages can cause emergencies. Rescue and emergency response operations must be initiated as swiftly as possible to save lives and mitigate the consequences of disasters. Accurate prediction of the precise time and locations of earthquakes is currently unattainable. Despite advancements in earthquake monitoring and geological research, monitoring and geological research, prediction remains challenging. However, monitoring systems can provide real-time earthquake information, such as magnitude and source location, for emergency response and warning purposes [222]. Earthquake evacuation models incorporate the impact of earthquakes on buildings and infrastructure, encompassing building collapse and road accessibility [32]. In addition, the potential impact of aftershocks must be considered. The impact of an earthquake lasts for a long time, necessitating both indoor and outdoor evacuations with the goal of reaching shelters. Earthquake evacuation studies typically employ both macroscopic and microscopic models to fully understand and simulate the evacuation process. Focusing on the overall earthquake evacuation situation, combined with GIS [223], factors such as large-scale population distribution [224], building density [225], and transportation networks [226] are considered. Microscopic models usually employ techniques such as agent-based modeling [227] to simulate the overall evacuation process by modeling individual behaviors. Individual factors such as walking speed [228], cognitive abilities [229], and panic factors [3] are taken into consideration. Environmental factors such as familiarity with the environment, guidance, and signage that can influence panic and herd behavior must be considered, given the large area affected by earthquakes.

Floods may affect entire cities or even larger areas. Establishing flood monitoring and early warning systems enables advanced knowledge of flood timing and scope, facilitating preparation for evacuation. Since extreme precipitation and flood events are affected by a variety of large-scale and small-scale weather systems, accurate predictions remain difficult. Although the urgency and degree of damage caused by floods are relatively low, their wide incidence and significant impact necessitate the prompt evacuation of people in low-lying and easily submerged areas to higher ground. Flood evacuation is often studied from a macroscopic perspective [124]. GIS technology, combined with topography [230], river networks [215], and other data, is utilized to establish flood spread models, perform flood dynamic simulations, and simulate flood paths, scopes and depths.

4. A general framework for future

The field of evacuation faces key challenges despite extensive research. Traditional evacuation plans, which frequently rely on statistical data, are hindered by the lack of detailed individual behavior models and real-time data, thereby diminishing their effectiveness in complex, urgent situations. Current path planning and guidance technologies are predominantly static, struggling to adapt to dynamic changes in emergencies. Furthermore, most evacuation strategies are designed for specific scenarios, lacking the necessary integration for seamless indoor-outdoor coordination. There is also a notable gap in effectively connecting different transportation modes during evacuations. Addressing these issues through fine-grained modeling, integrated multi-scenario approaches, and multimodal transportation coordination is essential for advancing evacuation planning and response [231].

1) Fine-grained simulation models

Fine-grained modeling, which focuses on accurately modeling individual behaviors and analyzing real-time data, is crucial for enhancing evacuation efficiency and safety. Traditional evacuation plans, which often rely on general statistical data [232], lack precision for complex emergencies. Effective fine-grained modeling requires integrating diverse data sources such as movement trajectories, social media, and sensor data to capture the nuanced behaviors influenced by individual characteristics, socio-cultural backgrounds, and psychological factors.

The challenge lies in capturing the diversity and uncertainty of crowd behaviors. Advanced data collection technologies, like IoT and sensor networks, are key to monitoring real-time crowd flow and extracting critical information such as congestion levels and preferred evacuation paths [53]. Combining these methods with machine learning and AI can develop precise models of individual behaviors, significantly improving evacuation planning accuracy. Moreover, fine-grained modeling must adapt to the dynamic nature of crowd behavior. Traditional static models fall short in reflecting real-time changes, thus making the integration of real-time data streams and dynamic models essential. This integration enables continuous model updates, providing real-time evacuation warnings and guidance. Additionally, integrating technologies such as virtual and augmented reality can simulate evacuation scenarios more realistically, enhancing decision-making training and plan optimization. This comprehensive approach to fine-grained modeling is crucial to significantly advancing evacuation strategies and responses.

Finally, these models must be adaptable to the unique characteristics of each type of disaster. In the context of earthquakes, the models must account for structural damage, predicting which areas might be blocked by debris or unsafe due to compromised buildings. For floods, the focus shifts to tracking water levels and flow rates and identifying safe evacuation routes as the situation evolves, particularly in urban areas where streets can become inundated. In fire scenarios, the model should incorporate real-time information on the spread of fire, smoke density, and heat levels, guiding individuals through the safest and quickest paths. Across all these disasters, the key is to understand and predict crowd behavior in response to rapidly changing conditions, ensuring that evacuation strategies are effective under chaotic and stressful conditions.

2) Multimodal Collaborative Emergency Evacuation

Traditional strategies employed in emergency evacuations frequently focus on specific transportation modes, neglecting the potential for integrating various modes for more efficient movement away from disaster areas. Multimodal collaborative evacuation addresses this gap by enabling transition between different transportation methods as needed, especially when certain modes become restricted or interrupted [233]. This approach not only enhances resilience but also improves the

overall efficiency and safety of evacuations. It involves careful planning and designing of evacuation routes, considering the unique characteristics and requirements of each transportation mode, and establishing effective coordination mechanisms.

Currently, most evacuation dispatching relies on static models based on predetermined road networks and traffic patterns [234]. However, real-life emergency scenarios demand dynamic traffic management and traffic volume prediction [235], which requires real-time monitoring and responsive decision-making. By employing intelligent transportation systems equipped with sensors, cameras, and GPS, real-time data on traffic congestion, evacuation routes, and vehicle availability can be analyzed. This enables adaptive evacuation guidance and traffic management, ensuring that optimal evacuation decisions are made under rapidly changing emergency conditions.

Tailored strategies for each disaster type can be obtained by adapting the concept of multimodal collaborative emergency evacuation to the unique challenges posed by earthquakes, floods, and fires. During earthquakes, structural damages often render roads unusable, necessitating alternatives such as aerial transport or off-road vehicles, guided by real-time seismic data. During floods, as roads become submerged, boats or amphibious vehicles are essential, with hydrological data directing these water-based evacuations. Fire scenarios, especially in urban areas, require a coordinated approach between pedestrian evacuation and public transportation, with fire tracking technologies determining safe pathways. The effectiveness of these diverse strategies in each disaster context depends on the real-time analysis of relevant data, using sensors, cameras, and GPS, ensuring that evacuation guidance is dynamically responsive to the specific conditions of each emergency situation.

3) Enhanced evacuation planning through GIS-BIM integration

Major emergencies and disasters often have widespread impacts, wherein both indoor and outdoor spaces can become hazardous areas requiring large-scale evacuation. Although integrating indoor and outdoor scenarios for unified emergency evacuation can effectively improve evacuation efficiency, practical challenges in implementation exist. Emergency situations in indoor and outdoor settings present different characteristics and constraints. Indoor spaces may be characterized by obstacles, narrow passageways, or staircases, which can impede crowd evacuation [101,236]. Visibility within indoor spaces may be limited, affecting individuals' perception of their surroundings and sense of direction [237]. In contrast, outdoor spaces may involve larger areas and complex geographic conditions, posing challenges for navigating and guiding crowd evacuation.

However, the development of GIS-BIM integration offers more accurate, efficient, and secure solutions for indoor-outdoor evacuation, thereby enhancing emergency response capabilities and effectiveness. BIM is capable of simulating real-time information about buildings, excelling in the detailed representation of interior components [238]. GIS supports spatial analysis and visual representation over large-scale spatial ranges [238]. The integration of GIS and BIM enables the unification of detailed representations of indoor spaces with macro-level analysis of outdoor spaces, facilitating data interoperability. By combining the BIM model of a building with GIS geospatial data, a comprehensive and accurate indoor-outdoor spatial model can be created [239]. This provides a more accurate data foundation for evacuation plans and emergency responses, enabling spatial intelligence and decision support for indoor-outdoor evacuation. By integrating real-time data and sensor information, the system can monitor the real-time positions of individuals, the status of building equipment, and other factors, providing accurate information for evacuation commands and decision-making. This helps guide decision-making during the evacuation process, such as selecting the optimal evacuation routes and coordinating rescue personnel.

5. Conclusion

This paper systematically reviews and categorizes the current state and challenges in emergency evacuation theory and methods, focusing on emphasizing evacuation simulation modeling and strategy optimization. The study highlights that current research primarily concentrates on evacuation simulation modeling, exploring methodologies such as cellular automata models, social force models, agent-based models, pedestrian flow models and network flow models, and providing insights into available simulation tools. This study also examines collective emergency evacuation strategies, discussing the application of these models in specific disaster scenarios. The complexity of crowd evacuation in major disasters, such as fires, earthquakes, and floods, is addressed, emphasizing the necessity of considering various factors, such as population dynamics and disaster types, to develop effective evacuation strategies. **The characteristics of different disaster accidents vary significantly**, and corresponding evacuation strategies and measures need to be formulated according to specific situations. For small-scale and highly urgent fires, it is necessary to avoid smoke threats and evacuate personnel safely to the exit as soon as possible. For large-scale and highly urgent earthquakes, personnel should be quickly evacuated to open spaces or shelters under sturdy structures to avoid the threat of collapsed buildings. For large-scale floods with relatively low urgency, the focus is on early warning and timely evacuation. Tailored evacuation strategies for specific disaster environments are achieved primarily through complex evacuation modeling and simulation techniques. **The key challenges identified include achieving accurate evacuation behavior modeling, ensuring seamless integration of indoor and outdoor evacuation, and coordinating multimodal collaborative evacuation procedures.** This research serves as a valuable reference for safety evacuation scholars, aiding in understanding and addressing the complexities of crowd evacuation during major disasters.

CRedit authorship contribution statement

Hai Sun: Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation.
Guorui Han: Writing – original draft, Resources, Methodology, Formal analysis.
Xiaowei Zhang: Writing – review & editing, Visualization.
Xuejing Ruan: Writing – review & editing, Visualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] PublicEM-DAT, [https://www.public.emdat.be/data\(2024\)](https://www.public.emdat.be/data(2024)).
- [2] M. Choi, S. Ji, J. Park, Risk assessments of crowds at bottlenecks through agent based modelling, *J. Korea. Geogr. Soc.* 57 (2022) 595–609, <https://doi.org/10.22776/kgs.2022.57.6.595>.
- [3] C.F. Liu, S.J. Liu, J.X. Zhang, et al., An optimal design method of emergency evacuation space in the high-density community after earthquake based on evacuation simulation, *Nat. Hazards* 116 (3) (2023) 2889–2915, <https://doi.org/10.1007/s11069-022-05789-8>, 2023.
- [4] B.Y. Li, J.M. Hou, Y.Y. Ma, et al., A coupled high-resolution hydrodynamic and cellular automata-based evacuation route planning model for pedestrians in flooding scenarios, *Nat. Hazards* 110 (1) (2022) 607–628, <https://doi.org/10.1007/s11069-021-04960-x>.
- [5] Y. Ahn, H. Choi, Development of a Real-time Safest Evacuation Route using Internet of Things and Reinforcement Learning in Case of Fire in a Building, *한국 안전학회지* 37 (2022) 97–105.
- [6] K. Deng, Q. Zhang, H. Zhang, P. Xiao, J. Chen, Optimal emergency evacuation route planning model based on fire prediction data, *Mathematics* 10 (2022), <https://doi.org/10.3390/math10173146>.
- [7] F.M. Khan, A. Kumar, H. Puppala, G. Kumar, R. Gupta, Projecting the criticality of COVID-19 transmission in India using GIS and machine learning methods, *J. Safety Sci. Resilience* 2 (2021) 50–62, <https://doi.org/10.1016/j.jnlssr.2021.05.001>.
- [8] X. Yang, Z. Wu, Y. Li, Difference between real-life earthquake escape panic and simulated exercises: Analysis of the video recordings of the 2008 Wenchuan earthquake, *Earthq. Res. China* 26 (2010) 242–250.
- [9] B. Bai, Y. Cao, X. Li, DTMM: Evacuation oriented optimized scheduling model for disaster management, *Comput. Commun.* 150 (2020) 661–671, <https://doi.org/10.1016/j.comcom.2019.11.049>.
- [10] Y. Jiang, Z. Li, S.L. Cutter, Social distance integrated gravity model for evacuation destination choice, *Int. J. Digit. Earth* 14 (2021) 1004–1018, <https://doi.org/10.1080/17538947.2021.1915396>.
- [11] Tsai, Y. Z. *Using AI for Improving Evacuation During Compound Disasters*, (2022).
- [12] V. Jurík, O. Uhlík, D. Snopková, et al., Analysis of the use of behavioral data from virtual reality for calibration of agent-based evacuation models, *Heliyon* 9 (3) (2023) e14275, <https://doi.org/10.1016/j.heliyon.2023.e14275>.
- [13] P. Murray-Tuite, B. Wolshon, Evacuation transportation modeling: An overview of research, development, and practice, *Transp. Res. Part C-Emerg. Technol.* 27 (2013) 25–45, <https://doi.org/10.1016/j.trc.2012.11.005>.
- [14] S.-K. Huang, M.K. Lindell, C.S. Prater, Who leaves and who stays? A review and statistical meta-analysis of hurricane evacuation studies, *Environ. Behav.* 48 (2016) 991–1029, <https://doi.org/10.1177/0013916515578485>.
- [15] Z. Kang, L. Zhang, K. Li, An improved social force model for pedestrian dynamics in shipwrecks, *Appl. Math. Comput.* 348 (2019) 355–362, <https://doi.org/10.1016/j.amc.2018.12.001>.
- [16] Y. Chen, S. Hu, H. Mao, W. Deng, X. Gao, Application of the best evacuation model of deep learning in the design of public structures, *Image Vis. Comput.* 102 (2020), <https://doi.org/10.1016/j.imavis.2020.103975>.
- [17] X. Yang, X. Yang, Q. Wang, Y. Kang, F. Pan, Guide optimization in pedestrian emergency evacuation, *Appl. Math. Comput.* 365 (2020), <https://doi.org/10.1016/j.amc.2019.124711>.
- [18] H. Liu, B. Xu, D. Lu, G. Zhang, A path planning approach for crowd evacuation in buildings based on improved artificial bee colony algorithm, *Appl. Soft. Comput.* 68 (2018) 360–376, <https://doi.org/10.1016/j.asoc.2018.04.015>.
- [19] J. Rocklov, H. Sjödin, A. Wilder-Smith, COVID-19 outbreak on the Diamond Princess cruise ship: estimating the epidemic potential and effectiveness of public health countermeasures, *J. Travel. Med.* 27 (2020), <https://doi.org/10.1093/jtm/taaa030>.
- [20] S. Tomer, G. Avishay, A. Bruria, Socializing in emergencies-A review of the use of social media in emergency situations, *Int. J. Inf. Manage* 35 (2015) 609–619, <https://doi.org/10.1016/j.ijinfomgt.2015.07.001>.
- [21] M. Haghighi, M. Sarvi, Crowd behaviour and motion: Empirical methods, *Transp. Res. Part B-Methodol.* 107 (2018) 253–293, <https://doi.org/10.1016/j.trb.2017.06.017>.
- [22] R.R. Thompson, D.R. Garfin, R.C. Silver, Evacuation from natural disasters: A systematic review of the literature, *Risk Anal.* 37 (2017) 812–839, <https://doi.org/10.1111/risa.12654>.
- [23] S. Gwynne, E.R. Galea, M. Owen, P.J. Lawrence, L. Filippidis, A review of the methodologies used in the computer simulation of evacuation from the built environment, *Build. Environ.* 34 (1999) 741–749, [https://doi.org/10.1016/S0360-1323\(98\)00057-2](https://doi.org/10.1016/S0360-1323(98)00057-2).
- [24] H. Sun, G.G. Huang, H. Jiang, et al., Study on simulation algorithm of crowdguided evacuation considering familiarity and panic effect, *J. Safety Sci. Technol.* 19 (2023) 5–12 (in Chinese).
- [25] J. Lin, Q. Ji, Z. Zhong, Modified social force model considering pedestrian characteristics and leaders, *Comput. Sci.* 49 (2022) 347–354.
- [26] H. Zhang, H. Wang, J. Nie, T. Wei, J. Wang, Structural equation model(SEM)of passengers helping behavior and its influencing factors in metro evacuation, *J. Safety Sci. Technol.* 19 (2023) 147–153.
- [27] O. Hosseini, M. Maghrebi, Risk of fire emergency evacuation in complex construction sites: Integration of 4D-BIM, social force modeling, and fire quantitative risk assessment, *Adv. Eng. Inf.* 50 (2021), <https://doi.org/10.1016/j.aei.2021.101378>.
- [28] Y. Li, M.Y. Chen, Z. Dou, et al., A review of cellular automata models for crowd evacuation, *Phys. A: Stat. Mech. Appl.* 526 (2019) 120752, <https://doi.org/10.1016/j.physa.2019.03.117>.
- [29] X. Chen, M. Treiber, V. Kanagaraj, H. Li, Social force models for pedestrian traffic - state of the art, *Transp. Res.* 38 (2018) 625–653, <https://doi.org/10.1080/01441647.2017.1396265>.
- [30] N. Siyam, O. Alqaryouti, S. Abdallah, Research issues in agent-based simulation for pedestrians evacuation, *IEE Access.* 8 (2020) 134435–134455, <https://doi.org/10.1109/ACCESS.2019.2956880>.
- [31] S.Q. Shan, X.S. Guo, Z.Y. Wei, et al., Simulation analysis of evacuation processes in a subway station based on multi-disaster coupling scenarios, *Int. J. Disast. Risk Reduct.* 96 (2023), <https://doi.org/10.1016/j.jidrr.2023.103998>.

- [32] M. De Iuliis, E. Battagazzorre, M. Domaneschi, G.P. Cimellaro, A.G. Bottino, Large scale simulation of pedestrian seismic evacuation including panic behavior, *Sustain. Cities*. 94 (2023), <https://doi.org/10.1016/j.scs.2023.104527>.
- [33] W.H. Li, M. Seike, A. Fujiwara, M. Chikaraishi, Structural equation modeling of negative emotion and walking behavior by smoke-filled model-scale tunnel experiments, *Saf. Sci.* 171 (2024), <https://doi.org/10.1016/j.ssci.2023.106394>.
- [34] R. Verma, Z.X. Lei, J.W. Xue, et al., How information heterogeneity influences traffic congestion during hurricane evacuation, in: *2021 IEEE International Intelligent Transportation Systems Conference (Itsc)*, 2021, pp. 1833–1838.
- [35] L. Lu, C.-Y. Chan, J. Wang, W. Wang, A study of pedestrian group behaviors in crowd evacuation based on an extended floor field cellular automaton model, *Transp. Res. Part D-Transp. Environ.* 81 (2017) 317–329, <https://doi.org/10.1016/j.trc.2016.08.018>.
- [36] Von Neumann, J. Theory of self-reproducing automata. Edited by Arthur W. Burks (1966).
- [37] X. Li, J. Wu, X. Li, *Theory of practical cellular automaton*, Springer, Germany, 2018.
- [38] M. Muramatsu, T. Irie, T. Nagatani, Jamming transition in pedestrian counter flow, *Phys. A: Stat. Mech. Appl.* 267 (1999) 487–498, [https://doi.org/10.1016/S0378-4371\(99\)00018-7](https://doi.org/10.1016/S0378-4371(99)00018-7).
- [39] C. Burstedde, K. Klauck, A. Schadschneider, J. Zittartz, Simulation of pedestrian dynamics using a two-dimensional cellular automaton, *Phys. A: Stat. Mech. Appl.* 295 (2001) 507–525, [https://doi.org/10.1016/S0378-4371\(01\)00141-8](https://doi.org/10.1016/S0378-4371(01)00141-8).
- [40] W.C. Liao, X.P. Zheng, L.S. Cheng, et al., Layout effects of multi-exit ticket-inspectors on pedestrian evacuation, *Saf. Sci.* 70 (2014) 1–8, <https://doi.org/10.1016/j.ssci.2014.04.015>.
- [41] A. Kirchner, A. Schadschneider, Simulation of evacuation processes using a bionics-inspired cellular automaton model for pedestrian dynamics, *Phys. A: Stat. Mech. Appl.* 312 (2002) 260–276, [https://doi.org/10.1016/S0378-4371\(02\)00857-9](https://doi.org/10.1016/S0378-4371(02)00857-9).
- [42] A. Varas, M.D. Cornejo, D. Mainemer, et al., Cellular automaton model for evacuation process with obstacles, *Phys. A: Stat. Mech. Appl.* 382 (2007) 631–642, <https://doi.org/10.1016/j.physa.2007.04.006>.
- [43] J.-W. Zhou, H. Kuang, M.-R. Liu, L.-J. Kong, Paired behavior effect on pedestrian evacuation dynamics, *ACTA Phys. Sin.* 58 (2009) 3001–3007, <https://doi.org/10.7498/aps.58.3001>.
- [44] R. Alizadeh, A dynamic cellular automaton model for evacuation process with obstacles, *Saf. Sci.* 49 (2011) 315–323, <https://doi.org/10.1016/j.ssci.2010.09.006>.
- [45] D.-M. Shi, B.-H. Wang, Evacuation of pedestrians from a single room by using snowdrift game theories, *Phys. Rev. E* 87 (2013), <https://doi.org/10.1103/PhysRevE.87.022802>.
- [46] L. Chen, T.Q. Tang, H.J. Huang, J.J. Wu, Z.Q. Song, Modeling pedestrian flow accounting for collision avoidance during evacuation, *Simul. Model. Pract. Theory.* 82 (2018) 1–11, <https://doi.org/10.1016/j.simpat.2017.12.011>.
- [47] X. Li, F. Guo, H. Kuang, Z. Geng, Y. Fan, An extended cost potential field cellular automaton model for pedestrian evacuation considering the restriction of visual field, *Physica A* 515 (2019) 47–56, <https://doi.org/10.1016/j.physa.2018.09.145>.
- [48] Y. Li, M. Chen, X. Zheng, Z. Dou, Y. Cheng, Relationship between behavior aggressiveness and pedestrian dynamics using behavior-based cellular automata model, *Appl. Math. Comput.* 371 (2020), <https://doi.org/10.1016/j.amc.2019.124941>.
- [49] K. Wang, Z. Fu, Y. Li, S. Qian, Influence of human-obstacle interaction on evacuation from classrooms, *Autom. Constr.* 116 (2020), <https://doi.org/10.1016/j.autcon.2020.103234>.
- [50] C. Chen, H. Sun, P. Lei, D. Zhao, C. Shi, An extended model for crowd evacuation considering pedestrian panic in artificial attack, *Physica A* 571 (2021), <https://doi.org/10.1016/j.physa.2021.125833>.
- [51] Y. Li, C. Lu, J. Jin, Simulation of a pediatric hospital in evacuation considering groups, *Simul. Model. Pract. Theory.* 107 (2021), <https://doi.org/10.1016/j.simpat.2020.102150>.
- [52] D.L. Gao, E.W.M. Lee, Y.Y. Lee, Integration of cumulative prospect theory in cellular automata model for building evacuation, *Int. J. Disast. Risk Reduct.* 74 (2022), <https://doi.org/10.1016/j.ijdrr.2022.102904>.
- [53] Y. Ji, W. Wang, M. Zheng, S. Chen, Real Time Building Evacuation modeling with an improved cellular automata method and corresponding IoT system implementation, *Buildings* 12 (2022), <https://doi.org/10.3390/buildings12060718>.
- [54] Y. Li, X. Yang, X. Gu, Y. Chen, Simulation of adult-child matching behavior: A case study of primary school in Linyi City, China, *Int. J. Mod. Phys. B* 36 (2022), <https://doi.org/10.1142/S0217979222501272>.
- [55] C. Guo, F. Huo, C. Li, Y. Li, An evacuation model considering the phototactic behavior of panic pedestrians under limited visual field, *Physica A* 615 (2023), <https://doi.org/10.1016/j.physa.2023.128602>.
- [56] Q. Xiao, J.Y. Li, Pedestrian evacuation model considering dynamic emotional update in direction perception domain, *Complexity*. 2021 (2021), <https://doi.org/10.1155/2021/5530144>.
- [57] R. Zhou, Y. Ou, W.C. Tang, Q. Wang, B. Yu, An Emergency evacuation behavior simulation method combines personality traits and emotion contagion, *IEEe Access.* 8 (2020) 66693–66706, <https://doi.org/10.1109/ACCESS.2020.2985987>.
- [58] L.-Y. Dong, L. Chen, X.-Y. Duan, Modeling and simulation of pedestrian evacuation from a single-exit classroom based on experimental features, *ACTA Phys. Sin.* 64 (2015), <https://doi.org/10.7498/aps.64.220505>.
- [59] F. Huo, Y. Li, C. Li, Y. Ma, An extended model describing pedestrian evacuation considering pedestrian crowding and stampede behavior, *Physica A* 604 (2022), <https://doi.org/10.1016/j.physa.2022.127907>.
- [60] L. Hou, J.G. Liu, X. Pan, Q. Guo, B.H. Wang, Simulation of pedestrian evacuation based on Jilin fire, *ACTA Phys. Sin.* 63 (2014), <https://doi.org/10.7498/aps.63.178902>.
- [61] Y. Song, K.F. Xie, W. Su, Mechanism and strategies of post-earthquake evacuation based on cellular automata model, *Int. J. Disast. Risk Reduct.* 34 (2019) 220–231, <https://doi.org/10.1016/j.ijdrr.2018.11.020>.
- [62] H. Sun, L.L. Hu, W.C. Shou, J. Wang, Self-organized crowd dynamics: research on earthquake emergency response patterns of drill-trained individuals based on GIS and multi-agent systems methodology, *Sensors* 21 (2021), <https://doi.org/10.3390/s21041353>.
- [63] J.X. Yi, S.L. Pan, Q. Chen, Simulation of pedestrian evacuation in stampedes based on a cellular automaton model, *Simul. Model. Pract. Theory.* 104 (2020), <https://doi.org/10.1016/j.simpat.2020.102147>.
- [64] P. Du, X. Liu, Z. Liu, X. Zhou, Simulation of subway evacuation under extreme rainfall based on cellular automata, *J. Safety Sci. Technol.* 18 (2022) 5–11.
- [65] C.C. Song, Q. Shao, P. Zhu, M. Dong, W.F. Yu, An emergency aircraft evacuation simulation considering passenger overtaking and luggage retrieval, *Reliab. Eng. Syst. Saf.* 229 (2023), <https://doi.org/10.1016/j.res.2022.108851>.
- [66] S.C. Cao, M.M. Wang, G. Zeng, X.L. Li, Simulation of crowd evacuation in subway stations under flood disasters, *IEEE Trans. Intell. Transp. Syst.* (2024), <https://doi.org/10.1109/TITS.2024.3372994>.
- [67] D. Helbing, Social force model for pedestrian dynamics, *Phys. Rev. E* 51 (1995) 4282–4286, <https://doi.org/10.1103/physreve.51.4282>.
- [68] D. Helbing, I. Farkas, T. Vicsek, Simulating dynamical features of escape panic, *Nature* 407 (2000) 487–490, <https://doi.org/10.1038/35035023>.
- [69] D. Helbing, L. Buzna, A. Johansson, T. Werner, Self-organized pedestrian crowd dynamics: Experiments, simulations, and design solutions, *Transp. Sci.* 39 (2005) 1–24.
- [70] A. Seyfried, B. Steffen, T. Lippert, Basics of modelling the pedestrian flow, *Phys. A: Stat. Mech. Appl.* 368 (2006) 232–238.
- [71] D.R. Parisi, M. Gilman, H. Moldovan, A modification of the social force model can reproduce experimental data of pedestrian flows in normal conditions, *Phys. A: Stat. Mech. Appl.* 388 (2009) 3600–3608.
- [72] G.A. Frank, C.O. Dorso, Room evacuation in the presence of an obstacle, *Phys. A: Stat. Mech. Appl.* 390 (2011) 2135–2145.
- [73] P. Saboia, S. Goldenstein, Crowd simulation: applying mobile grids to the social force model, *Vis. Comput.* 28 (2012) 1039–1048.
- [74] Gao, Y., Luh, P. B., Zhang, H. & Chen, T. in *2013 IEEE International Conference on Automation Science and Engineering (CASE)*. 747–751 (IEEE).
- [75] G. Köster, F. Tremel, M. Gödel, Avoiding numerical pitfalls in social force models, *Phys. Rev. E* 87 (2013) 063305.
- [76] O.M. Alia, M.M. Shuaib, A harmony search algorithm for the reproduction of experimental data in the social force model, *J. Appl. Math.* 2014 (2014).
- [77] X. Yang, H. Dong, X. Yao, Passenger distribution modelling at the subway platform based on ant colony optimization algorithm, *Simul. Model. Pract. Theory.* 77 (2017) 228–244, <https://doi.org/10.1016/j.simpat.2017.03.005>.
- [78] A.M. Ibrahim, M. Saifullah, M.R.M. Romlay, et al., Hybrid Social Force-Fuzzy Logic Evacuation Simulation Model for Multiple Exits, in: *2019 7th International Conference on Mechatronics Engineering (ICOM)*, 2019, pp. 140–144.
- [79] D.S. Stuart, M.S. Sharifi, K.M. Christensen, et al., Crowds involving individuals with disabilities: Modeling heterogeneity using fractional order potential fields and the social force model, *Phys. A: Stat. Mech. Appl.* 514 (2019) 244–258.
- [80] I.M. Sticco, G.A. Frank, F.E. Cornes, C.O. Dorso, A re-examination of the role of friction in the original Social Force Model, *Saf. Sci.* 121 (2020) 42–53.
- [81] Y. Han, W. Wang, Q. Fan, Z. Hu, Optimization of building evacuation exits based on genetic algorithm and pedestrian evacuation model, *Comput. Eng. Appl.* 56 (2020) 254–261.
- [82] J. Wei, Y.Y. Guo, L. You, An improved crowd evacuation model for a multi-exit environment, *China Safety Sci. J. (CSSJ)* 30 (2020) 139–145 (in Chinese).
- [83] X. Liu, Y. Zhang, X. Cai, Simulation study of personnel evacuation in highway tunnel based on modified social force model by introducing into ventilation force, *J. Safety Sci. Technol.* 17 (2021) 43–47.
- [84] Q. Mao, R. Yu, C. Mao, Research on multi-exit evacuation model based on improved butterfly algorithm and social force model, *J. Safety Sci. Technol.* 18 (2022) 12–18.
- [85] A. García, D. Hernández-Delfín, D.-J. Lee, M. Ellero, Limited visual range in the Social Force Model: Effects on macroscopic and microscopic dynamics, *Phys. A: Stat. Mech. Appl.* 612 (2023) 128461, <https://doi.org/10.1016/j.physa.2023.128461>.
- [86] M.F. Li, Y.X. Zhao, L.R. He, W.X. Chen, X.F. Xu, The parameter calibration and optimization of social force model for the real-life 2013 Ya'an earthquake evacuation in China, *Saf. Sci.* 79 (2015) 243–253, <https://doi.org/10.1016/j.ssci.2015.06.018>.
- [87] G. Bernardini, M. Postacchini, E. Quagliarini, et al., A preliminary combined simulation tool for the risk assessment of pedestrians' flood-induced evacuation, *Environ. Model. Softw.* 96 (2017) 14–29.
- [88] X. Lu, Z. Yang, G.P. Cimellaro, Z. Xu, Pedestrian evacuation simulation under the scenario with earthquake-induced falling debris, *Saf. Sci.* 114 (2019) 61–71.
- [89] J.X. Zhou, S. Li, G.Z. Nie, X.W. Fan, C.X. Xia, Developing a revised social force model for pedestrians' earthquake emergency evacuation, *Geomat. Nat. Hazards Risk* 11 (2020) 335–356, <https://doi.org/10.1080/19475705.2020.1724202>.
- [90] J.B. Zhang, J. Zhu, P. Dang, et al., An improved social force model (ISFM)-based crowd evacuation simulation method in virtual reality with a subway fire as a

- case study, *Int. J. Digit. Earth*. 16 (2023) 1186–1204, <https://doi.org/10.1080/17538947.2023.2197261> (in Chinese).
- [91] G.C. Zhong, G.F. Zhai, W. Chen, Optimization of shelter location based on a combined static/dynamic two-stage optimization methodology: A case study in the Central Urban Area of Xinyi City, China, *ISPRS. Int. J. Geoinf.* 11 (2022), <https://doi.org/10.3390/ijgi11040262>.
- [92] X.W. Shao, J.J. Feng, J.H. Wang, Y. Wang, Research on emergency evacuation behavior rules of pedestrians under fire and explosion accidents in chemical parks considering social relationships, *J. Loss. Prev. Process. Ind.* 87 (2024), <https://doi.org/10.1016/j.jlp.2024.105244>.
- [93] Macal, C. M. & North, M. J. in *Proceedings of the Winter Simulation Conference*, 2005. 14 pp.
- [94] S. Sharma, K. Ogunlana, D. Scribner, J. Grynowicki, Modeling human behavior during emergency evacuation using intelligent agents: A multi-agent simulation approach, *Inf. Syst. Front.* 20 (2018) 741–757, <https://doi.org/10.1007/s10796-017-9791-x>.
- [95] H. Lim, H.Y.S. Lee, J.H. Hwang, Multi-agent simulation on staff evacuation behavior in elderly nursing home fire emergencies, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13020400>.
- [96] E. Bonabeau, Agent-based modeling: Methods and techniques for simulating human systems, *Proc. Nat. Acad. Sci.* 99 (2002) 7280–7287, <https://doi.org/10.1073/pnas.082080899>.
- [97] X. Pan, C.S. Han, K. Dauber, K.H. Law, A multi-agent based framework for the simulation of human and social behaviors during emergency evacuations, *AI. Soc.* 22 (2007) 113–132, <https://doi.org/10.1007/s00146-007-0126-1>.
- [98] X.W. Chen, J. Wang, Iop, in: *2nd International Conference on Advanced Electronic Materials, Computers and Materials Engineering (AEMCME)*, 2019.
- [99] S. Hassanpour, A.A. Rassafi, Agent-based simulation for pedestrian evacuation behaviour using the affordance concept, *KSCIE J. Civil Eng.* 25 (2021) 1433–1445, <https://doi.org/10.1007/s12205-021-0206-7>.
- [100] A. Gobbin, R. Khosravi, A. Bardenhagen, Emergency evacuation simulation of commercial aircraft Influence of body- and behaviour parameters, *SN. Appl. Sci.* 3 (2021), <https://doi.org/10.1007/s42452-021-04295-z>.
- [101] Y. Bao, F. Huo, An agent-based model for staircase evacuation considering agent's rotational behavior, *Phys. A: Stat. Mech. Appl.* 572 (2021) 125923.
- [102] L.A. Cotfas, C. Delcea, L.D. Iancu, C. Ioană, C. Ponsiglione, Large event halls evacuation using an agent-based modeling approach, *IEEe Access.* 10 (2022) 49359–49384, <https://doi.org/10.1109/ACCESS.2022.3172285>.
- [103] C. Wang, C. Zhu, K. Xiao, Y. Tang, H. Zhen, Visualization of emergency evacuation physical behavior under multi-agent decision-making, *Appl. Sci.-Basel* 13 (2023), <https://doi.org/10.3390/app13095509>.
- [104] E. Mas, S. Koshimura, F. Imamura, et al., Recent Advances in Agent-Based Tsunami Evacuation Simulations: Case Studies in Indonesia, Thailand, Japan and Peru, *Pure Appl. Geophys.* 172 (2015) 3409–3424, <https://doi.org/10.1007/s00024-015-1105-y>.
- [105] W. Yang, Y. Hu, C. Hu, M. Yang, An Agent-Based Simulation of deep foundation pit emergency evacuation modeling in the presence of collapse disaster, *Symmet.-Basel* 10 (2018), <https://doi.org/10.3390/sym10110581>.
- [106] Z.Q. Wang, J. Huang, H.M. Wang, J.L. Kang, W.W. Cao, Analysis of flood evacuation process in vulnerable community with mutual aid mechanism: an agent-based simulation framework, *Int. J. Environ. Res. Public Health* 17 (2020), <https://doi.org/10.3390/ijerph17020560>.
- [107] M. Shirvani, G. Kesserwani, P. Richmond, Agent-based simulator of dynamic flood-people interactions, *J. Flood. Risk. Manage* 14 (2021) e12695, <https://doi.org/10.1111/jfr3.12695>.
- [108] S. Grajdura, S. Borjigin, D. Niemeier, Fast-moving fire wildfire evacuation simulation, *Transp. Res. Part D-Transp. Environ.* 104 (2022), <https://doi.org/10.1016/j.trd.2022.103190>.
- [109] K. Chapuis, P. Minh-Duc, A. Brugière, et al., Exploring multi-modal evacuation strategies for a landlocked population using large-scale agent-based simulations, *Int. J. Geogr. Inf. Sci.* 36 (2022) 1741–1783, <https://doi.org/10.1080/13658816.2022.2069774>.
- [110] W. Song, X. Xu, B.-H. Wang, S. Ni, Simulation of evacuation processes using a multi-grid model for pedestrian dynamics, *Phys. A: Stat. Mech. Appl.* 363 (2006) 492–500, <https://doi.org/10.1016/j.physa.2005.08.036>.
- [111] R.Y. Guo, H.J. Huang, A mobile lattice gas model for simulating pedestrian evacuation, *Physica A* 387 (2008) 580–586, <https://doi.org/10.1016/j.physa.2007.10.001>.
- [112] Q. Liu, J. Liang, M. Li, Y. Fu, Study on crowd evacuation behaviors based on multi-agent and cellular automata technology, *Comput. Simul.* 31 (2014) 328–332.
- [113] Y.M. Chen, C. Wang, H. Li, et al., Cellular automaton model for social forces interaction in building evacuation for sustainable society, *Sustain. Cities. Soc.* 53 (2020) 101913, <https://doi.org/10.1016/j.scs.2019.101913>.
- [114] H. Ren, Y. Yan, F. Gao, Variable guiding strategies in multi-exits evacuation: Pursuing balanced pedestrian densities, *Appl. Math. Comput.* (2021) 397, <https://doi.org/10.1016/j.amc.2021.125965>.
- [115] Y. Liu, G. Huang, W. Chen, Improved social force model based on enhancing psych behavioral heterogeneity, *J. Syst. Simul.* 35 (2023) 1120–1130.
- [116] D. Helbing, H.J. Herrmann, M. Schreckenberg, D.E. Wolf, *Traffic and granular flow '99: social, traffic, and granular dynamics*, Springer Science & Business Media, 2012.
- [117] Kretz, T. An overview of fundamental diagrams of pedestrian dynamics. <https://doi.org/10.13140/RG.2.2019>.
- [118] L. Henderson, The statistics of crowd fluids, *Nature* 229 (1971) 381–383.
- [119] R.L. Hughes, A continuum theory for the flow of pedestrians, *Transp. Res. Part B: Methodol.* 36 (2002) 507–535.
- [120] R. Narain, A. Golas, S. Curtis, M.C. Lin, Aggregate dynamics for dense crowd simulation, *ACM Trans. Graph.* 28 (2009), <https://doi.org/10.1145/1618452.1618468>.
- [121] M. Twarogowska, P. Goatin, R. Duvinneau, in: *Conference on Pedestrian and Evacuation Dynamics 2014 (PED 2014)* Vol. 2, 2014, pp. 477–485.
- [122] M. Twarogowska, P. Goatin, R. Duvinneau, Macroscopic modeling and simulations of room evacuation, *Appl. Math. Model.* 38 (2014) 5781–5795, <https://doi.org/10.1016/j.apm.2014.03.027>.
- [123] G.-Y. Wu, M. Mizuno, The numerical analysis of mass evacuation in Taipei 101 with control volume model, *Simul. Model. Pract. Theory.* 96 (2019) 101937, <https://doi.org/10.1016/j.simpat.2019.101937>.
- [124] S. Maity, S. Sundar, A coupled model for macroscopic behavior of crowd in flood induced evacuation, *Physica A* 607 (2022), <https://doi.org/10.1016/j.physa.2022.128161>.
- [125] Y.-Q. Jiang, S.-G. Zhou, Y.-L. Duan, X.-Q. Huang, A viscous continuum model with smoke effect for pedestrian evacuation, *Physica A* 621 (2023), <https://doi.org/10.1016/j.physa.2023.128792>.
- [126] W.J. Zhu, R.Y. Zhao, H. Zhang, et al., Panic-pressure conversion model from microscopic pedestrian movement to macroscopic crowd flow, *J. Comput. Nonlinear. Dyn.* 18 (12) (2023) 121001, <https://doi.org/10.1115/1.4063505>.
- [127] L.R. Ford, D.R. Fulkerson, Maximal flow through a network, *Canad. J. Math.* 8 (1956) 399–404.
- [128] J.A. Huertas, D. Duque, E. Segura-Duran, R. Akhavan-Tabatabaei, A.L. Medaglia, Evacuation dynamics: a modeling and visualization framework, *OR Spect.* 42 (2020) 661–691, <https://doi.org/10.1007/s00291-019-00548-x>.
- [129] R.L.A. Francis, Uniformity principle" for evacuation route allocation, *J. Res. Natl. Bur. Stand.* (1977) 86 (1981) 509–513, <https://doi.org/10.6028/jres.086.023>.
- [130] N. Kamiyama, N. Katoh, A. Takizawa, An efficient algorithm for evacuation problem in dynamic network flows with uniform arc capacity, *IEICE Trans. Inf. Syst.* E89D (2006) 2372–2379, <https://doi.org/10.1093/ietisy/e89-d.8.2372>.
- [131] Y. Shin, I. Moon, Robust building evacuation planning in a dynamic network flow model under collapsible nodes and arcs, *Socioecon. Plann. Sci.* 86 (2023), <https://doi.org/10.1016/j.seps.2022.101455>.
- [132] H.N. Nath, U. Pyakurel, T.N. Dhamala, S. Dempe, Dynamic network flow location models and algorithms for quickest evacuation planning, *J. Ind. Manag. Optimiz.* 17 (2021) 2943–2970, <https://doi.org/10.3934/jimo.2020102>.
- [133] X. Zheng, Y. Chen, L. Jin, M. Zhang, L. Xiang, Evacuation crowd diversion model for road intersection, *China Safety Sci. J. (CSSJ)* 28 (2018) 176–182.
- [134] Y. Cui, H. Xu, K. Gong, A diversion routing optimization model for urban evacuation planning, *Nat. Hazards* 116 (2023) 2399–2416, <https://doi.org/10.1007/s11069-022-05771-4>.
- [135] L. Chen, P. Lin, Quickest evacuation of flood detention basin, *Eng. J. Wuhan Univ./Wuhan Daxue Xuebao* 45 (2012) 468–471.
- [136] W. Lu, F. Wang, L. Bu, L. Liu, Game approach to vulnerability analysis of evacuation highway networks, *J. Transp. Eng. Part A-Syst.* 145 (2019), <https://doi.org/10.1061/JTEPBS.0000263>.
- [137] N. Zhang, H. Huang, B.N. Su, J.L. Zhao, Analysis of dynamic road risk for pedestrian evacuation, *Physica A* 430 (2015) 171–183, <https://doi.org/10.1016/j.physa.2015.02.082>.
- [138] T. Wang, J. Zhou, Y. Jiang, et al., Flood refuge and migration model based on network flow, *J. Nat. Disast.* 25 (2016) 56–64.
- [139] J.G. Jin, Y.F. Shen, H. Hu, Y.Q. Fan, M.J. Yu, Optimizing underground shelter location and mass pedestrian evacuation in urban community areas: A case study of Shanghai, *Transp. Res. Part A-Policy Pract.* 149 (2021) 124–138, <https://doi.org/10.1016/j.tra.2021.04.009>.
- [140] M. Kinader, C.Y. Ma, S. Gwynne, M. Amos, N. Bénichou, Where drills differ from evacuations: A case study on Canadian buildings, *Saf. Sci.* 135 (2021), <https://doi.org/10.1016/j.ssci.2020.105114>.
- [141] Y. Ding, L. Yang, F. Weng, Z. Fu, P. Rao, Investigation of combined stairs elevators evacuation strategies for high rise buildings based on simulation, *Simul. Model. Pract. Theory.* 53 (2015) 60–73, <https://doi.org/10.1016/j.simpat.2015.01.004>.
- [142] X. Long, X. Zhang, B. Lou, Numerical simulation of dormitory building fire and personnel escape based on Pyrosim and Pathfinder, *J. China. Inst. Eng.* 40 (2017) 257–266, <https://doi.org/10.1080/02533839.2017.1300072>.
- [143] A. Kallianiotis, D. Papakonstantinou, V. Arvelaki, A. Benardos, Evaluation of evacuation methods in underground metro stations, *Int. J. Disast. Risk Reduct.* 31 (2018) 526–534, <https://doi.org/10.1016/j.ijdrr.2018.06.009>.
- [144] Y. Tang, X. Biao, F. Ma, Research on fire evacuation of shopping malls based on pathfinder, *J. Shenyang Jianzhu Univ.. Nat. Sci.* 35 (2019) 858–866.
- [145] J. Qin, C. Liu, Q. Huang, Simulation on fire emergency evacuation in special subway station based on Pathfinder, *Case Stud. Therm. Eng.* 21 (2020) 100677.
- [146] W. Yan, R. Zhang, J. Liu, Y. Zhang, W. Sun, Study on fire evacuation of university library based on pathfinder, *J. Shenyang Jianzhu Univ.. Nat. Sci.* 37 (2021) 627–633.
- [147] M.F. Xiao, X.H. Zhou, X.X. Pan, et al., Simulation of emergency evacuation from construction site of prefabricated buildings, *Sci. Rep.* 12 (1) (2022) 2732, <https://doi.org/10.1038/s41598-022-06211-w>.
- [148] Z. Liu, X. Gu, R. Hong, Fire Protection and evacuation analysis in underground interchange tunnels by integrating BIM and numerical simulation, *Fire-Switzerland* 6 (2023), <https://doi.org/10.3390/fire6040139>.
- [149] S. Lee, Y.-J. Son, J. Jin, An integrated human decision making model for evacuation scenarios under a BDI framework, *ACM Trans. Model. Comput. Simul.* 20 (2010), <https://doi.org/10.1145/1842722.1842728>.

- [150] X. Zhao, Y. Hu, Research on fuzzy modeling and simulation of evacuees' behavior, *China Safety Sci. J.(CSSJ)* 25 (2015) 56–61.
- [151] K. Zhang, S. Yang, L. He, Y. Long, Research on emergency evacuation of passengers with luggage from bus station, *China Safety Sci. J.(CSSJ)* 27 (2017) 30–35.
- [152] M. Shi, E.W.M. Lee, Y. Ma, A novel grid-based mesoscopic model for evacuation dynamics, *Physica A* 497 (2018) 198–210, <https://doi.org/10.1016/j.physa.2017.12.139>.
- [153] H.X. Chen, H. Li, H.B. Zhang, Y.L. Wang, simulation analysis of urban rail transit station evacuation considering line of sight obstruction, in: 5th International Conference on Information Science, Computer Technology and Transportation (ISCTT), 2020, pp. 645–649.
- [154] J. Tang, Y.L. Hu, Lee, in: 33rd Chinese Control and Decision Conference (CCDC), 2021, pp. 2763–2767.
- [155] F. Chasanah, H. Sakakibara, Implication of mutual assistance evacuation model to reduce the volcanic risk for vulnerable society: insight from Mount Merapi, Indonesia, *Sustainability* 14 (2022), <https://doi.org/10.3390/su14138110>.
- [156] H. Lian, S. Zhang, G. Li, Y. Zhang, Pedestrian simulation on evacuation behavior in teaching building of primary school emergencies and optimized design, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13071747>.
- [157] Z.C. Zhang, H.B. Li, E.W.M. Lee, et al., Application of Minecraft in the study of evacuation dynamics under fire emergency conditions, *Phys. A: Stat. Mech. Appl.* 624 (2023) 128935, <https://doi.org/10.1016/j.physa.2023.128935>.
- [158] H.M. Yang, D.L. Zhao, P.C. Liu, et al., Study on safety evacuation strategy with coupling of stairs and elevators in high-rise building, *J. Safety Sci. Technol.* 15 (2019) 169–174 (in Chinese).
- [159] C. Wang, Y. Song, Fire evacuation in metro stations: modeling research on the effects of two key parameters, *Sustainability* 12 (2020), <https://doi.org/10.3390/su12020684>.
- [160] L. Filippidis, H. Xie, E.R. Galea, P.J. Lawrence, Exploring the potential effectiveness of dynamic and static emergency exit signage in complex spaces through simulation, *Fire Saf. J.* 125 (2021), <https://doi.org/10.1016/j.firesaf.2021.103404>.
- [161] M.T. Liu, Y.F. Wang, W. Zhu, M.M. Yin, J.C. Zheng, simulation of exit choice behaviour in evacuation considering the obstacles and pedestrian distribution, in: Proceedings of 2021 IEEE 12th International Conference on Software Engineering and Service Science (ICSESS), 2021, pp. 228–234.
- [162] J.F. Chen, Z. Long, L. Wang, et al., Fire evacuation strategy analysis in long metro tunnels, *Saf. Sci.* 147 (2022) 105603, <https://doi.org/10.1016/j.ssci.2021.105603>.
- [163] N. Mu, G. Zha, X. Xu, Analysis on influence of diversion railing setting on personnel evacuation of subway island platform, *J. Safety Sci. Technol.* 18 (2022) 199–203.
- [164] L. Chang, X. He, W. Song, Research on steel structure building evacuation based on empirical formula method and SIMULEX, *Acta Scientiarum Naturalium Universitatis Nankaiensis* 49 (2016) 21–28.
- [165] C.-H. Shao, P.-C. Shao, F.-M. Kuo, Stampede events and strategies for crowd management, *J. Disast. Res.* 14 (2019) 949–958, <https://doi.org/10.20965/jdr.2019.p0949>.
- [166] Y. Xu, S. Liao, M. Liu, Simulation and assessment of fire evacuation modes for long underwater vehicle tunnels, *Fire Technol.* 55 (2019) 729–754, <https://doi.org/10.1007/s10694-018-0798-8>.
- [167] E.-y. Ko, C.H.A. Jaesang, A Study on the Efficiency of Evacuation Guidance and Non-evacuation Guidance in Case of Fire, *Int. J. Adv. Cult. Technol.* 8 (2020) 243–247.
- [168] Z.Y. Lin, Z. Liu, Z.P. Fang, et al., Review of pedestrian and traffic evacuation simulation software, *China Safety Sci. J.(CSSJ)* 32 (2022) 100–110.
- [169] M. Kobes, I. Helsloot, B. de Vries, J.G. Post, Building safety and human behaviour in fire: A literature review, *Fire Saf. J.* 45 (2010) 1–11, <https://doi.org/10.1016/j.firesaf.2009.08.005>.
- [170] P.E. Adam St John, FDS Modeling Analysis, ATF Fire Research Laboratory, 2021.
- [171] L. Yang, W. Fang, R. Huang, Z. Deng, Occupant evacuation model based on cellular automata in fire, *Chine. Sci. Bull.* 47 (2002) 1484–1488.
- [172] W. Yuan, K.H. Tan, A model for simulation of crowd behaviour in the evacuation from a smoke-filled compartment, *Phys. A: Stat. Mech. Appl.* 390 (2011) 4210–4218.
- [173] Y.-C. Peng, C.-I. Chou, Simulation of pedestrian flow through a “T” intersection: A multi-floor field cellular automata approach, *Comput. Phys. Commun.* 182 (2011) 205–208.
- [174] X. Zhang, X. Li, G. Hadjisophocleous, A probabilistic occupant evacuation model for fire emergencies using Monte Carlo methods, *Fire Saf. J.* 58 (2013) 15–24.
- [175] X. Guo, J. Chen, S. You, J. Wei, Modeling of pedestrian evacuation under fire emergency based on an extended heterogeneous lattice gas model, *Phys. A: Stat. Mech. Appl.* 392 (2013) 1994–2006.
- [176] N. Mu, W.-G. Song, X.-X. Qi, W. Lu, S.-C. Cao, Simulation of evacuation in a twin bore tunnel: analysis of evacuation time and egress selection, *Procedia Eng.* 71 (2014) 333–342, <https://doi.org/10.1016/j.proeng.2014.04.048>.
- [177] Y. Zheng, B. Jia, X.-G. Li, R. Jiang, Evacuation dynamics considering pedestrians' movement behavior change with fire and smoke spreading, *Saf. Sci.* 92 (2017) 180–189.
- [178] V. Kodur, S. Venkatachari, M. Naser, Egress parameters influencing emergency evacuation in high-rise buildings, *Fire Technol.* 56 (2020) 2035–2057.
- [179] Y. Wang, J. Ma, Y. Liu, Y. Bai, L. Xu, Optimal exit choice during highway tunnel evacuations based on the fire locations, *PLoS. One* 16 (2021), <https://doi.org/10.1371/journal.pone.0256523>.
- [180] S. Gao, C. Chang, Q. Liu, M. Zhang, F. Yu, Study on the optimization for emergency evacuation scheme under fire in university building complex, *Heliyon* 9 (2023), <https://doi.org/10.1016/j.heliyon.2023.e14277>.
- [181] A. Tucker, K.L. Marsh, T. Gifford, et al., The effects of information and hazard on evacuee behavior in virtual reality, *Fire Saf. J.* 99 (2018) 1–11.
- [182] Y. Zhou, Y. Pang, F. Chen, Y. Zhang, Three-dimensional indoor fire evacuation routing, *ISPRS. Int. J. GeoInf.* 9 (2020) 558.
- [183] H.-H. Yen, C.-H. Lin, H.-W. Tsao, Time-aware and temperature-aware fire evacuation path algorithm in IoT-enabled multi-story multi-exit buildings, *Sensors* 21 (2020) 111.
- [184] M. Amini, D.R. Sanderson, D.T. Cox, A.R. Barbosa, N. Rosenheim, Methodology to incorporate seismic damage and debris to evaluate strategies to reduce life safety risk for multi-hazard earthquake and tsunami, *Nat. Hazards* (2023), <https://doi.org/10.1007/s11069-023-05937-8>.
- [185] K. Ertugay, S. Argyroudis, H.S. Duzgun, Accessibility modeling in earthquake case considering road closure probabilities: A case study of health and shelter service accessibility in Thessaloniki, Greece, *Int. J. Disast. Risk Reduct.* 17 (2016) 49–66, <https://doi.org/10.1016/j.ijdr.2016.03.005>.
- [186] E. Mas, A. Suppasri, F. Imamura, S. Koshimura, Agent-based simulation of the 2011 Great East Japan earthquake/tsunami evacuation: An integrated model of tsunami inundation and evacuation, *J. Nat. Disast. Sci.* 34 (2012) 41–57.
- [187] G. Bernardini, M. D'Orazio, E. Quagliarini, L. Spalazzi, An agent-based model for earthquake pedestrians' evacuation simulation in urban scenarios, *Transp. Res. Procedia* 2 (2014) 255–263.
- [188] M.-L. Xiao, Y. Chen, M.-J. Yan, L.-Y. Ye, B.-Y. Liu, Simulation of household evacuation in the 2014 Ludian earthquake, *Bull. Earthq. Eng.* 14 (2016) 1757–1769.
- [189] J.L. Aguado, T.M. Ferreira, P.B. Lourenço, The use of a large-scale seismic vulnerability assessment approach for masonry façade walls as an effective tool for evaluating, managing and mitigating seismic risk in historical centers, *Int. J. Arch. Herit.* 12 (2018) 1259–1275.
- [190] S. Santarelli, G. Bernardini, E. Quagliarini, M. D'Orazio, New indices for the existing city-centers streets network reliability and availability assessment in earthquake emergency, *Int. J. Arch. Herit.* 12 (2018) 153–168.
- [191] M. Xiao, Y. Zhang, H. Zhu, The mechanism of emergency occupants' evacuation from seismic responses of building, *Nat. Hazards* 96 (2019) 669–692.
- [192] Z.N. Feng, V.A. González, M. Trotter, et al., How people make decisions during earthquakes and post-earthquake evacuation: Using verbal protocol analysis in immersive virtual reality, *Saf. Sci.* 129 (2020) 104837.
- [193] J. Yang, H. Zhan, L. Chen, Z. Dou, Planning of emergency evacuation routes in densely populated urban areas during earthquakes, *J. Tsinghua Univ. Sci. Technol.* 62 (2022) 70–76.
- [194] A.M. El-Maissi, S.A. Argyroudis, M.M. Kassem, L.V. Leong, F.M. Nazri, An integrated framework for the quantification of road network seismic vulnerability and accessibility to critical services, *Sustainability* 14 (2022), <https://doi.org/10.3390/su141912474>.
- [195] L. Alcada-Almeida, L. Tralhao, L.-s. Santos, J. Coutinho-Rodrigues, A multiobjective approach to locate emergency shelters and identify evacuation routes in urban areas, *Geogr. Anal.* 41 (2009) 9–29, <https://doi.org/10.1111/j.1538-4632.2009.00745.x>.
- [196] Y. Bayram, B.C. Tansel, H. Yaman, Compromising system and user interests in shelter location and evacuation planning, *Transp. Res. Part B: Methodol.* 72 (2015) 146–163.
- [197] Esposito Amideo, A. & Scaparra, M. P. in *International Conference on Optimization and Decision Science*. 567-576 (Springer).
- [198] Yamamoto, K. & Shimura, Y. in *2017 Joint Urban Remote Sensing Event (JURSE)*. 1-4 (IEEE).
- [199] J. Zhang, X. Zhao, L. Bai, Y. Wang, F. Zheng, Space requirements and routes optimization of earthquake-proof evacuation on University Campus in earthquake-prone zones, *J. Jilin Univ. Earth Sci. Ed.* 52 (2022) 305–314.
- [200] B. Wei, X. Dong, Q. Tan, W. Qi, X. Guo, Analysis of the earthquake emergency evacuation route based on the capacity of shelters, *J. Seismol. Res.* 45 (2022) 141–149.
- [201] M. Keykhaei, N.N. Samany, M. Jelokhani-Niaraki, S. Zlatanova, A situation-aware emergency evacuation (SAEE) model using multi-agent-based simulation for crisis management after earthquake warning, *Geo-Spat. Inf. Sci.* (2023), <https://doi.org/10.1080/10095020.2023.2270017>.
- [202] Cramer, F. Evacuation bottlenecks in flooding events (Marcel Thielmann).
- [203] S.P. Simonovic, S. Ahmad, Computer-based model for flood evacuation emergency planning, *Nat. Hazards* 34 (2005) 25–51, <https://doi.org/10.1007/s11069-004-0785-x>.
- [204] R.Y. Guo, H.J. Huang, Logit-based exit choice model of evacuation in rooms with internal obstacles and multiple exits, *Chine. Phys. B* 19 (2010) 030501.
- [205] E. Du, X. Cai, Z. Sun, B. Minsker, Exploring the role of social media and individual behaviors in flood evacuation processes: An agent-based modeling approach, *Water. Resour. Res.* 53 (2017) 9164–9180.
- [206] Y. Li, B.S. Hu, D. Zhang, et al., Flood evacuation simulations using cellular automata and multiagent systems - a human-environment relationship perspective, *Int. J. Geogr. Inf. Sci.* 33 (2019) 2241–2258, <https://doi.org/10.1080/13658816.2019.1622015>.
- [207] H.-K. Lee, W.-H. Hong, Y.-H. Lee, Experimental study on the influence of water depth on the evacuation speed of elderly people in flood conditions, *Int. J. Disast. Risk Reduct.* 39 (2019) 101198.
- [208] Y. Zheng, X.-G. Li, B. Jia, R. Jiang, Simulation of pedestrians' evacuation dynamics with underground flood spreading based on cellular automaton, *Simul.*

- Model. Pract. Theory. 94 (2019) 149–161, <https://doi.org/10.1016/j.simpat.2019.03.001>.
- [209] H. Nakanishi, S. Wise, Y. Suenaga, E. Manley, Simulating emergencies with transport outcomes Sim (SETOSim): Application of an agent-based decision support tool to community evacuation planning, *Int. J. Disast. Risk Reduct.* 49 (2020) 101657.
- [210] C. Huang, D. Su, L. Yang, G. Lei, W. Huang, Study of pedestrian evacuation in subway station considering the process of flood spreading, *J. Rail Way Sci. Eng.* 18 (2021) 2865–2872.
- [211] F. Southworth, S.M. Chin, Network evacuation modelling for flooding as a result of dam failure, *Environ. Plann. A* 19 (1987) 1543–1558.
- [212] C.E. Dunn, D. Newton, Optimal routes in GIS and emergency planning applications, *Area* (1992) 259–267.
- [213] T. Yamada, A network flow approach to a city emergency evacuation planning, *Int. J. Syst. Sci.* 27 (1996) 931–936.
- [214] S. Jamrussri, Y. Toda, Available flood evacuation time for high-risk areas in the middle reach of Chao Phraya River Basin, *Water*. (Basel) 10 (2018), <https://doi.org/10.3390/w10121871>.
- [215] D. Yin, GIS network analysis technology for flood evacuation routes, *J. Yangtze River Sci. Res. Inst.* 33 (2016) 38–41.
- [216] S. Park, G. Lee, J.O. Kim, Flood evacuation mapping using a time-distance cartogram, *ISPRS. Int. J. Geoinf.* 9 (2020), <https://doi.org/10.3390/ijgi9040207>.
- [217] O.-E. Chelariu, C. Iatu, I. Minea, A GIS-based model for flood shelter locations and pedestrian evacuation scenarios in a rural mountain catchment in Romania, *Water*. (Basel) 14 (2022), <https://doi.org/10.3390/w14193074>.
- [218] Z. Yan, Y. Wang, L. Chao, Simulation study on fire and evacuation of super high-rise commercial building, *Case Stud. Therm. Eng.* 52 (2023), <https://doi.org/10.1016/j.csite.2023.103519>.
- [219] W. Lei, A. Li, C. Tai, The effect of different fire source locations on the environment of a subway station, in: *10th International Symposium on Heating, Ventilation and Air Conditioning*, (ISHVAC)2017 Vol. 205, 2017, pp. 3721–3726.
- [220] Y. Zheng, B. Jia, X.-G. Li, R. Jiang, Evacuation dynamics considering pedestrians' movement behavior change with fire and smoke spreading, *Saf. Sci.* 92 (2017) 180–189, <https://doi.org/10.1016/j.ssci.2016.10.009>.
- [221] Y.M. Chen, C. Wang, J.B.H. Yap, et al., Fire evacuation process using both elevators and staircases for aging people: Simulation case study on personnel distribution in high-rise nursing home, *Discrete Dyn. Nat. Soc.* 2020 (3) (2020) 1–14, <https://doi.org/10.1155/2020/5365126>.
- [222] Allen, R. M. & Melgar, D. Earthquake early warning: advances, scientific challenges, and societal needs. *Ann. Rev. Earth Planet. Sci.*, 47, 361–388 (2019).
- [223] J.H. Xu, X.Z. Yin, D.C. Chen, J.W. An, G.Z. Nie, Multi-criteria location model of earthquake evacuation shelters to aid in urban planning, *Int. J. Disast. Risk Reduct.* 20 (2016) 51–62, <https://doi.org/10.1016/j.ijdrr.2016.10.009>.
- [224] X.J. Zhao, W. Xu, Y.J. Ma, et al., Relationships between evacuation population size, earthquake emergency shelter capacity, and evacuation time, *Int. J. Disast. Risk Sci.* 8 (2017) 457–470, <https://doi.org/10.1007/s13753-017-0157-2>.
- [225] C. Chen, L. Cheng, Influence of urban built environment on the safety and efficiency of residential seismic evacuation behavior, *Journal of Safety and Environment* 22 (2022) 3350–3360.
- [226] V. Bayram, H. Yaman, A stochastic programming approach for shelter location and evacuation planning, *Rairo-Oper. Res.* 52 (2018) 779–805, <https://doi.org/10.1051/ro/2017046>.
- [227] G. Bernardini, M. D'Orazio, E. Quagliarini, L. Spalazzi, in: *Conference on Pedestrian and Evacuation Dynamics 2014 (PED 2014) Vol. 2*, 2014, pp. 255–263.
- [228] M. D'Orazio, L. Spalazzi, E. Quagliarini, G. Bernardini, Agent-based model for earthquake pedestrians' evacuation in urban outdoor scenarios: Behavioural patterns definition and evacuation paths choice, *Saf. Sci.* 62 (2014) 450–465, <https://doi.org/10.1016/j.ssci.2013.09.014>.
- [229] T. Kono, M. Okuno, K. Yamaura, Regional differences in cognitive dissonance in evacuation behavior at the time of the 2011 Japan earthquake and tsunami, *Nat. Hazards* 111 (2022) 139–162, <https://doi.org/10.1007/s11069-021-05047-3>.
- [230] M.N. He, C. Chen, F.F. Zheng, et al., An efficient dynamic route optimization for urban flooding evacuation based on Cellular Automata, *Comput. Environ. Urban Syst.* 87 (2021) 101622, <https://doi.org/10.1016/j.compenvurbysys.2021.101622>.
- [231] J. Jiang, A review on emergency evacuation methods for major sudden disasters and accidents, *Geom. Inf. Sci. Wuhan Univ.* 46 (2021) 1498–1518.
- [232] C.S. Jiang, Y. Ling, C. Hu, et al., Numerical simulation of emergency evacuation of a subway station: A case study in Beijing, *Archit. Sci. Rev.* 52 (2009) 183–193, <https://doi.org/10.3763/asre.2008.0047>.
- [233] Z. Xu, Q. Bai, Y. Shao, A. Hu, Z. Dong, A review on passenger emergency evacuation from multimodal transportation hubs, *J. Traff. Transp. Eng.-Engl. Ed.* 9 (2022) 591–607, <https://doi.org/10.1016/j.jtte.2022.02.001>.
- [234] S. Xu, Study on fire evacuation route of underground shopping mall sharing exit with subway, *Xi'an University of Architecture and Technology*, 2020.
- [235] Y.Z. Song, X.Y. Wang, G. Wright, et al., Traffic volume prediction with segment-based regression kriging and its implementation in assessing the impact of heavy vehicles, *IEEE Trans. Intell. Transp. Syst.* 20 (2019) 232–243, <https://doi.org/10.1109/TITS.2018.2805817>.
- [236] A. Varas, M.D. Cornejo, D. Mainemer, et al., Cellular automaton model for evacuation process with obstacles, *Phys. A: Stat. Mech. Appl.* 382 (2007) 631–642, <https://doi.org/10.1016/j.physa.2007.04.006>.
- [237] X. Li, F. Guo, H. Kuang, Z. Geng, Y. Fan, An extended cost potential field cellular automaton model for pedestrian evacuation considering the restriction of visual field, *Phys. A: Stat. Mech. Appl.* 515 (2019) 47–56, <https://doi.org/10.1016/j.physa.2018.09.145>.
- [238] S. Xu, Research on BIM/GIS based emergency evacuation method in metro station, *Shanghai Jiao Tong University*, 2019.
- [239] Y.Z. Song, X.Y. Wang, Y. Tan, et al., Trends and opportunities of BIM-GIS integration in the architecture, engineering and construction industry: A review from a spatio-temporal statistical perspective, *ISPRS. Int. J. Geoinf.* 6 (2017) 397.